Chiral generating pairs in finite simple groups: a complete assembly of the positive theorem

Human Compression project

Abstract

We assemble a proof of the positive existence theorem of Leemans and Liebeck for finite nonabelian simple groups. The proof combines explicit alternating generators, finite word certificates, and involution-counting arguments with all parameter ranges accounted for. The classification and geometric reconstruction theorems used as antecedents are stated explicitly. Several auxiliary assertions in the original proof are replaced by corrected arguments, without changing its positive conclusion. No new existence theorem or classification is claimed.

The full proofs and certificates used below are retained in their exact component editions. The accepted general orthogonal counting argument and the separately accepted D_4^+ cover are joined explicitly in Section 4.4. Historical pending-review language in a frozen component records its earlier author stage; the acceptance and dependency records bound to this assembly state the completed interfaces. This manuscript establishes the positive principal theorem, with the source boundaries specified in Section 5.

1 The exact theorem and its geometric meaning

The fixed source is arXiv:1603.07713v3, whose running header is dated 18 May 2016 and whose internal date is 6 November 2018. Its numbered principal theorem is the following positive statement [1].

Theorem 1 (Leemans–Liebeck)**.** *Let G be a finite nonabelian simple group which is not isomorphic to A_7, $\mathrm{PSL}_2(q)$, $\mathrm{PSL}_3(q)$ or $\mathrm{PSU}_3(q)$. There are $x, t \in G$ such that*

$$G = \langle x, t\rangle, \qquad |t| = 2, \qquad \nexists \alpha \in \mathrm{Aut}(G): \quad x^\alpha = x^{-1}, \quad t^\alpha = t.$$

The source asks only that no *involutory* automorphism have the specified action. These formulations are equivalent for a generating pair: the square of such an automorphism fixes both generators and hence is the identity. The identity itself cannot invert x, since $|x| > 2$. Indeed, a group generated by two involutions is dihedral, whereas G is nonabelian simple. The automorphism group in the theorem is the full $\mathrm{Aut}(G)$; restricting to inner automorphisms would not suffice.

Lemma 2. *Every pair in Theorem 1 satisfies the rank-three rotation intersection conditions and yields an abstract chiral polyhedron with full automorphism group G.*

Proof. Set $y = x^{-1}t$. Then $xy = t$, so $(xy)^2 = 1$, and $G = \langle x, y\rangle$. An element of $\langle x\rangle \cap \langle y\rangle$ commutes with both generators, and therefore is central in G. This intersection is trivial. Neither x nor y has order at most two: otherwise G would be cyclic or generated by two involutions. Thus the orders are at least three.

We now invoke the retained Schulte–Weiss rank-three reconstruction and regularity criterion [2], in the form used in the source: these rotation data reconstruct a directly regular or chiral polyhedron; the directly regular alternative is precisely the existence of the reversing automorphism in Theorem 1. Its absence gives chirality, and the reconstructed full automorphism group is G. □

Corollary 3. *Every group in Theorem 1 is the full automorphism group of an abstract chiral polyhedron.*

This uses geometric reconstruction as an explicit antecedent, rather than claiming a new construction theorem for abstract polytopes. Equivalence of marked rotation data means a group isomorphism carrying the distinguished generators to their counterparts. No bijection between *literal labelled pairs* and unmarked polyhedra is asserted here. In particular, no enumerative count is deduced by forgetting orientation, identifying enantiomorphs, or dividing the number of pairs by an unstated symmetry factor.

2 Two sufficient criteria

The following elementary arguments explain the two forms of certificate used throughout the proof. Write $i_2(H)$ for the number of elements of order exactly two, and $\nu_2(H) = i_2(H) + 1$ for the number of solutions of $h^2 = 1$.

Lemma 4 (Word criterion)**.** *If $G = \langle x, t\rangle$, $|t| = 2$, and a word w satisfies $|w(x,t)| \neq |w(x^{-1},t)|$, no automorphism of G inverts x and fixes t.*

Proof. An automorphism with those generator images sends one word value to the other, whereas automorphisms preserve element order. □

All forty-seven explicit finite certificates used below employ $w(x,t) = xtx^2tx^3t$: twelve sporadic cases and thirty-five Lie parameter entries. These are parameter certificates, not forty-seven distinct abstract groups, since some small classical names coincide.

Lemma 5 (Counting criterion)**.** *Let G be finite and centreless, and fix $x \in G$ of order greater than two. Suppose $\mathcal{H}$ is a family of actual proper subgroups covering every proper subgroup containing x. If*

$$i_2(G) > \sum_{\substack{\alpha \in \mathrm{Aut}(G) \\ |\alpha|=2,\ x^\alpha = x^{-1}}} i_2(C_G(\alpha)) + \sum_{H \in \mathcal{H}} i_2(H),$$

then an involution t satisfies the conclusion of Theorem 1.

Proof. The second sum bounds involutions that generate a proper subgroup with x. The first bounds involutions fixed by any involutory reverser. Choose t outside their union. Any reverser fixing this generating partner has square one, as observed above, and is already excluded. □

Overlaps among these sets only enlarge the bound, so no disjointness is required. A list of *types* or conjugacy classes of subgroups is not yet $\mathcal{H}$: the number of actual subgroups containing x must be proved. The cyclic normalizer must also be covered or separately charged. These points are paid explicitly by the acquired components.

A convenient bound for the first sum is the following. If $G \leq J \lhd \mathrm{Aut}(G)$ and $C_J(x) = T$, each outer coset contains at most $|T|$ reversers. Only square-one elements of $\mathrm{Aut}(G)/J$ can be images of involutory reversers. If their number is κ and every relevant fixed group has at most μ involutions, the first sum is at most $\kappa|T|\mu$. This does not assume that all reversers are conjugate. When passing through a kernel K, we use $\nu_2(H) \leq |K|\nu_2(H/K)$, including the identity before taking fibres. No algebraic dimension estimate is substituted for a finite kernel order.

3 Alternating and sporadic groups

For A_n, $n \geq 8$, the complete alternating component [3] proves generation and excludes the full reversing automorphism group for the source's explicit pairs:

n even	$x = (2,3,\ldots,n)$	$t = (1,2)(3,4)$,
n odd	$x = (1,2,\ldots,n)$	$t = (1,2)(3,6)$.

Its proof is uniform in n; the finite controls only check displayed identities and do not prove an unbounded theorem by sampling. The standard alternating-group automorphism theorem and the stated primitive permutation-group antecedent remain retained. The lower alternating names $A_5 \cong \mathrm{PSL}_2(4)$ and $A_6 \cong \mathrm{PSL}_2(9)$ lie in the excluded isomorphism families, as does A_7 by name.

The twenty-six sporadic groups are divided into the disjoint sets

$$\mathcal{S}_{12} = \{M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, J_1, J_2, J_3, HS, He, Suz, McL\},$$
$$\mathcal{S}_{14} = \{J_4, Co_1, Co_2, Co_3, Fi_{22}, Fi_{23}, Fi'_{24}, M, B, Th, Ly, \mathrm{O'N}, HN, Ru\}.$$

For $\mathcal{S}_{12}$, the exact permutation certificates [4] check membership in each identified named group, full generation, and the unequal word orders of Lemma 4. The M_{23}, M_{24} source permutations are retained literally. Other witnesses may have different polyhedron types from the source examples; their role is existence. Named representation identifications and simplicity are retained inputs, not inferred from an order match alone.

For $\mathcal{S}_{14}$, the complete counting component [5] chooses x of an indicated odd prime order p. Counting conjugates of a maximal subgroup M containing x gives the bound

$$B_G(x) = |C_G(x)| \sum_{[M]} \frac{i_p(M) i_2(M)}{|M|}.$$

The sum includes every maximal-subgroup conjugacy class, including repeated character-table names for distinct classes. The inverter term is zero if x and x^{-1} are not conjugate in the full automorphism group; otherwise it is at most $|C_{\mathrm{Aut}(G)}(x)|\mu$. The component supplies a rigorous μ from complete maximal lists in G or its full automorphism group, checking the required centralizer indices in the four outer-index-two cases. Its fourteen strict rational inequalities prove Lemma 5. Ordinary character tables, class fusions, maximal-subgroup completeness, and the named automorphism-group identifications are explicit retained classification premises. Thus all twenty-six sporadic cases are covered without using any historical whole-map census or its numerical total.

4 The Lie families and all exceptional parameters

We use the standard classification of finite nonabelian simple groups, including its low-rank isomorphisms, to pass from these families to all simple groups. In the classical notation below, groups are projective where the centre is nontrivial. Orthogonal odd dimension has odd q; in characteristic two the corresponding simple groups are symplectic.

4.1 Finite parameter certificates

The finite component [6] supplies every simple entry in the source's two finite exception lists, together with its three exceptional Lie cases. For $\epsilon = \pm$, $L^+ = \mathrm{PSL}$ and $L^- = \mathrm{PSU}$, these are

$$L_4^\epsilon(2), L_4^\epsilon(3), L_4^\epsilon(4), L_5^\epsilon(2), L_6^\epsilon(2), \quad \mathrm{PSp}_4(3), \mathrm{Sp}_4(4), \mathrm{Sp}_6(2), \mathrm{PSp}_6(3), \mathrm{Sp}_8(2),$$
$$\Omega_7(3), \Omega_8^\epsilon(2), P\Omega_8^+(3), \Omega_{10}^\epsilon(2),$$
$$L_5^\epsilon(3), L_6^\epsilon(3), L_8^\epsilon(2), \mathrm{Sp}_6(4), \mathrm{PSp}_8(3), \mathrm{Sp}_8(4), \Omega_{12}^\epsilon(2),$$
$$G_2(3), G_2(4), {}^2F_4(2)'.$$

There are $21 + 11 + 3 = 35$ parameter entries. The source also prints $\mathrm{Sp}_4(2) \cong S_6$, which is not simple and is not a positive obligation. The overlaps $\mathrm{PSL}_4(2) \cong A_8$ and $\mathrm{PSU}_4(2) \cong \mathrm{PSp}_4(3)$ are preserved as duplicate parameter routes, not counted as new abstract groups. Each positive certificate verifies its actual named-constructor interface, membership, full generation and Lemma 4; a candidate-producing search is not the correctness premise.

4.2 Exceptional Lie type

The complete exceptional component [7] treats

$$E_8, E_7, E_6, {}^2E_6, F_4, G_2, {}^3D_4, {}^2F_4, {}^2G_2, {}^2B_2$$

in every simple parameter not covered by the finite list. The small identifications $G_2(2)' \cong \mathrm{PSU}_3(3)$ and ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$ belong to excluded families and are not converted into positive cases. The Tits group is supplied by the finite certificate, not by treating ${}^2F_4(2)$ as simple.

The component proves regularity of the chosen torus elements by a root-lattice argument, including its small finite cases. It bounds reversers by outer cosets, identifies the admissible algebraic involution classes using the reviewed flag-variety argument, and charges the cyclic normalizer. Explicit class orders give the involution lower bounds; translated polynomials with nonnegative coefficients and positive constant terms prove all uniform margins. The residual fixed parameters are checked exactly. Torus existence, complete maximal-overgroup types, class centralizers and the uniform Lawther–Liebeck–Seitz upper theorem [16] remain named antecedents. In particular, the proof does not invoke Seitz's theorem outside its field-size hypotheses or infer a finite lower constant from an asymptotic estimate.

4.3 Linear, unitary and symplectic type

The elementary classical lower component [8] proves

G	$i_2(G) > q^D/4$
$\mathrm{PSL}_n(q), \mathrm{PSU}_n(q),\ n \geq 4$	$D = \lfloor n^2/2 \rfloor$
$\mathrm{PSp}_{2m}(q),\ m \geq 2$	$D = m^2 + m$
$\Omega_{2m+1}(q),\ m \geq 3$	$D = m^2 + m$
$P\Omega^{\epsilon}_{2m}(q),\ m \geq 4$	$D = m^2 - 1.$

It counts actual involution families with their scalar losses; no maximal-subgroup classification is needed for these lower estimates. For linear and unitary groups, the complete upper component [9] constructs the Singer or $(n-1)+1$ element, proves its adjoint torus centralizer and the single possible reversing coset, and bounds the actual fixed groups, including characteristic-two nonalternating forms. It then proves the strict counting inequality for every parameter outside the displayed finite list. The exact source Table 4 linear/unitary cover is explicitly retained as a named classification assertion, with its Bereczky–Bamberg–Penttila provenance and the published author erratum. The later acquisition of an older full BP preprint does not retroactively turn that frozen component into an independent proof of the classification.

For symplectic groups, the complete component [10] uses the full Singer cover in Bubboloni–Spiga–Weigel Lemma 5.1 [17], including the odd-characteristic unitary overgroup omitted from the source table. Trace forms classify the natural reversers; the exceptional graph-field coset in degree four and even characteristic is separately included. Actual finite kernels and the degree-four estimates give strict inequalities in all remaining ranks and fields. The finite list supplies all small excluded parameters. Thus both components meet exactly the hypotheses of Lemma 5.

4.4 Orthogonal type

The torus component [11] supplies elements with their actual spinor, Dickson and projective scalar conditions. In odd dimension use a squared primitive norm-one multiplier on a negative hyperplane; in minus even dimension use a primitive multiplier for even q and its square for odd q; in plus dimension use primitive multipliers on the negative $(2m-2)+2$ decomposition. The connected projective conformal centralizer orders are respectively

$$q^m + 1, \qquad q^m + 1, \qquad (q^{m-1} + 1)(q + 1).$$

The odd and even fixed-group components [12, 13] pay the finite centralizers of natural reversing automorphisms, including nonsquare multipliers and unipotent kernels. In particular the full-rank even characteristic degree-eight bound is $\nu_2(C_G(\alpha)) \leq q^{11}$.

The orthogonal counting component [15] uses the complete BSW overgroup statements in odd dimension, minus dimension, and plus dimension at least ten. It proves the multiplicities of actual field structures, includes a bound for the cyclic normalizer, and establishes the uniform inequalities. Its remaining parameters are exactly those already present in the finite list, except for the separately treated dimension-eight range.

In plus dimension eight, the complete D4 specialization [14] proves, for every primitive choice on the negative $6+2$ decomposition, that the only maximal types are its decomposition stabilizer, at most two unitary extension normalizers, and the triality B_3 subgroups. It derives the cover from the complete published BHRD Table 8.50 [18]; the table's number of conjugacy classes is not used as an element-containing multiplicity. The independent triality argument in [15] chooses the primitive components so that all B_3 overgroups are absent for even $q \geq 4$ and odd $q \geq 7$. At $q = 5$ it counts the actual six possible line stabilizers, not the source table's asserted two. The same argument controls the additional inverter cosets. With the exact degree-eight lower estimates, the displayed strict inequalities prove the claim at all these fields. The finite certificates pay plus type at $q = 2, 3$ and minus type at $q = 2$. This is the precise interface at which the complete D4 cover discharges the conditional hypothesis of the counting manuscript; no later result is used to justify that cover.

5 Completion and source boundaries

Proof of Theorem 1. By the retained classification of finite simple groups, every group under the hypothesis is alternating, sporadic or of Lie type, with the standard small identifications described above. The alternating proof covers all eligible alternating groups. The two disjoint sporadic rosters cover all twenty-six sporadics. The finite list, exceptional component and three classical routes cover every eligible Lie parameter. Each route establishes the same full-automorphism generating-pair conditions, either by Lemma 4 or Lemma 5. There is no assumption that an auxiliary table is itself a theorem about an unverified new element. This proves the group-theoretic assertion; Lemma 2 proves Corollary 3. □

The manuscript establishes this *positive* theorem. On source page 3, nonexistence for $\mathrm{PSL}_2(q)$ is attributed to Macbeath and Singerman, and nonexistence for A_7 to the earlier atlas computation. These are historical antecedents, not deductions of our positive construction. The corresponding $\mathrm{PSL}_3(q), \mathrm{PSU}_3(q)$ statement is explicitly announced there as forthcoming work [19], not proved in the fixed source. We do not promote that announcement into a proved converse. The source's introductory classification of regular polyhedron groups, and its final almost-simple Suzuki and PSL_2-socle statements, likewise remain attributed antecedents. The general almost-simple extension is a question, not an additional conclusion.

The Section 6 counts for $M_{11}, M_{12}, M_{22}, J_1, J_2$, the cited all-types O'Nan enumeration, and example types not realized by our replacement witnesses are auxiliary historical statements. Our theorem requires their existence conclusion only. No new exhaustive map count, faithfulness of an unrelated historical census, or claim to recover every ancillary numerical example is made. In particular the O'Nan existence argument does not import the disputed total from the separate L046 source.

Several source proof interfaces have been replaced: the omitted normalizer term is covered; outer inverters are not assumed to have one class; small field hypotheses are respected; the symplectic unitary overgroup is included; orthogonal scalar losses and triality multiplicities are corrected. The source's special estimate $i_2(\mathrm{PSU}_4(q)) > q^8/2$ is false for $q \equiv 3 \pmod 4$, as established by the exact formula in [8]. The valid quarter bound and sharper finite calculations

suffice here. These are repairs to supporting assertions; they do not refute the positive principal theorem.

Attribution and retained costs. The theorem, principal strategy and alternating pairs belong to Leemans–Liebeck and their cited predecessors. The full component proofs and exact certificates listed below are part of this assembly, not merely status summaries. The classification of finite simple groups, Schulte–Weiss reconstruction, named finite-group constructions and character tables, complete maximal-subgroup classifications, stated automorphism structure, and the cited Lie-type involution inputs remain explicit borrowed results. They are not claimed to have been reproved or formally verified.

The root agent authored the alternating, twelve-sporadic, sporadic counting, exceptional, orthogonal torus, odd fixed-group, triality and orthogonal counting work. Publication_recovery authored the finite Lie certificates, classical lower bounds, linear/unitary, symplectic, even fixed-group and D4 cover components and this assembly. The root supplied the finite candidates and several organizing observations; l044_resume supplied the exceptional flag-variety amendment, which has its own separate review. The two relative-field arguments were developed in parallel and are not counted as independent reviews of each other. f003_resume supplied the bounded order-21/63 note and distinctly reviewed the general counting/multiplicity component. Source-contract assistance by l044_resume is recorded separately from mathematical acceptance. Earlier bounded proofs, their distinct reviews, and all retained classification costs remain attached to their exact editions. No novelty, new question resolution, Lean verification or new measurement is asserted.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, fixed fourteen-page source.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, DIMACS Series 4 (1991), 493–516; retained rank-three reconstruction.

[3] *Alternating groups generated by a cycle and two transpositions*, complete acquired L045 alternating component, version 2, with distinct review and root adoption.

[4] *Chirality detected by the order of one word*, complete acquired twelve-sporadic component, version 2, with distinct review and adoption.

[5] *Counting involutions that generate a chiral polyhedron*, complete acquired fourteen-sporadic component, version 1, with distinct review.

[6] *Finite Lie-group witnesses for chiral polyhedra*, complete acquired thirty-five-parameter component, version 1, with distinct review.

[7] *Involutions outside the reversal and subgroup obstructions*, exceptional component, version 2, including the separately reviewed flag-variety amendment and complete arithmetic certificates.

[8] *Explicit involution classes for the classical part of L045*, version 1, with exact unitary correction and distinct review.

[9] *Torus reversers and explicit bounds for the linear and unitary branches*, version 1, with distinct review.

[10] *Explicit symplectic reverser and overgroup bounds*, version 2, with distinct review.

[11] *Orthogonal tori and their projective orders*, version 1, with distinct review.

[12] *Bounds for orthogonal fixed groups in odd characteristic*, version 2, with distinct review.

[13] *Finite kernels of full-rank orthogonal inverters in characteristic two*, version 1, with distinct root review.

[14] *Maximal overgroups of a full negative* $6 + 2$ *torus element*, version 1; complete type specialization distinctly reviewed, shared multiplicity interface separately routed through the counting review.

[15] *Counting overgroups of an orthogonal torus element* and *Choosing an orthogonal torus element to separate the triality modules*, complete orthogonal counting component, version 1; with its separate distinct review and the independently accepted complete D_4^+ cover.

[16] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), 393–464, especially Proposition 1.3.

[17] D. Bubboloni, P. Spiga and Th. Weigel, *Normal* 2*-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Lemmas 5.1, 6.1, 7.3 and 8.1.

[18] J. N. Bray, D. F. Holt and C. M. Roney-Dougal, *The Maximal Subgroups of the Low-Dimensional Finite Classical Groups*, LMS Lecture Note Series 407, Cambridge University Press, 2013, Main Theorem 2.1.1 and Table 8.50.

Alternating groups generated by a cycle and two transpositions

Human Compression project

We give complete elementary generation and chirality arguments for the two families stated in Section 5 of Leemans–Liebeck [1]. The source attributes these particular permutations to Gareth Jones. The families, and the existence conclusion, are not new results of this note. Our purpose is to supply their complete proof through one elementary generation lemma, so that no search over the alternating groups is required.

1 A generation lemma

For a finite set U, let $\mathrm{Alt}(U)$ act trivially outside U.

Lemma 1. *Suppose $|U| \geq 3$ and T is a set of three points meeting U. Then $\langle \mathrm{Alt}(U), \mathrm{Alt}(T) \rangle = \mathrm{Alt}(U \cup T)$.*

Proof. First suppose T has two points a, b in U and one new point d. Conjugating $(a\ b\ d)$ by $\mathrm{Alt}(U)$ gives a 3-cycle supported on $\{i, j, d\}$ for every pair of distinct points $i, j \in U$. When $|U| \geq 4$, this follows because $\mathrm{Alt}(U)$ is transitive on ordered pairs: choose any permutation taking one ordered pair to the other and, if its sign is wrong, compose with a transposition of two other points. When $|U| = 3$, its cyclic action is transitive on the three unordered pairs; inverses give either orientation of the resulting 3-cycles. Together with $\mathrm{Alt}(U)$, these are all 3-cycles on $U \cup \{d\}$. They generate its alternating group. Indeed an even permutation is a product of an even number of transpositions; pairing them expresses it as a product of 3-cycles, since two intersecting transpositions have a 3-cycle as product and, for distinct points, $(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d)$.

If $T \cap U = \{a\}$, write $T = \{a, d, e\}$ and choose distinct $b, c \in U \setminus \{a\}$. Conjugating $(a\ b\ c)$ by a suitable element of $\mathrm{Alt}(T)$ gives a 3-cycle on $\{d, b, c\}$. The first case therefore supplies $\mathrm{Alt}(U \cup \{d\})$. The original set T now meets that enlarged set in two points, so a second application supplies $\mathrm{Alt}(U \cup T)$. The case $T \subseteq U$ is immediate. All generators are even, giving equality rather than merely containment. □

Lemma 2. *A primitive permutation group on a finite set Ω containing a 3-cycle contains $\mathrm{Alt}(\Omega)$.*

Proof. Take the supports of all conjugates of that 3-cycle. Regard these triples as the edges of a hypergraph on Ω. Its connected components form a partition preserved by the group, because conjugation permutes the triples. There is a component of size at least three. Transitivity and primitivity therefore force the hypergraph to be connected. Start with one triple and successively adjoin a triple meeting the union already obtained until all vertices are included. Such an ordering exists by connectedness. Each conjugate generates the alternating group of its three-point support. Lemma 1 consequently shows that the subgroup generated by all these conjugates contains $\mathrm{Alt}(\Omega)$. □

This is the familiar 3-cycle case of the primitive-group generation criterion. The argument above includes its proof; it is not a new classification theorem.

2 The two families

We use the standard fact $\mathrm{Aut}(\mathrm{Alt}(n)) \cong \mathrm{Sym}(n)$ for $n \geq 8$, where a permutation acts by conjugation. This automorphism theorem is retained as an external group-theoretic premise; we do not claim to prove it here.

Theorem 3. *For even $n \geq 8$, put*

$$x = (2\ 3\ \cdots\ n), \qquad t = (1\ 2)(3\ 4).$$

For odd $n \geq 9$, put

$$x = (1\ 2\ \cdots\ n), \qquad t = (1\ 2)(3\ 6).$$

In either case $\langle x, t\rangle = \mathrm{Alt}(n)$, and no automorphism of $\mathrm{Alt}(n)$ both inverts x and fixes t.

Proof. Let $G = \langle x, t\rangle$. Both permutations are even, so $G \leq \mathrm{Alt}(n)$. In the even case, x is transitive on all points except 1, and t moves 1 into that orbit. Hence G is transitive. In any G-invariant partition, the part containing 1 is fixed by x. It therefore contains either none or all of $\{2, \ldots, n\}$. The partition is trivial, proving primitivity. The conjugate of t obtained by applying x^3 to its point labels is $(1\ 5)(6\ 7)$. Its product with t has a 3-cycle on $\{1, 2, 5\}$ and the two disjoint transpositions $(3\ 4), (6\ 7)$. Squaring this product leaves a single 3-cycle. Lemma 2 now gives $G = \mathrm{Alt}(n)$.

In the odd case, x already makes G transitive. In a nontrivial invariant partition, every part would have the same size, a proper divisor of the odd number n, and hence size at least three. If t moved one part to another, it would move every point of at least two parts, and thus at least six points. This contradicts its support of size four. Thus t would fix every part, forcing 1 and 2 into the same part. Applying powers of x puts every consecutive pair in a common part and forces the whole set into one part, a contradiction. Thus G is primitive. Here the conjugate obtained by applying x^2 to the labels of t is $(3\ 4)(5\ 8)$. Its product with t consists of a 3-cycle on $\{3, 4, 6\}$ and the disjoint transpositions $(1\ 2), (5\ 8)$. Its square is a 3-cycle, so again Lemma 2 gives $G = \mathrm{Alt}(n)$.

It remains to exclude the unwanted automorphism. By the retained automorphism theorem it is induced by a permutation h. In the even case, inversion of x makes h fix its unique fixed point 1. If h also fixes t under conjugation, it must fix 2, since $(1\ 2)$ is the unique transposition of t meeting 1. Reversing the cycle x while fixing 2 sends 3 to n and 4 to $n - 1$. It cannot preserve the other transposition $(3\ 4)$ for $n \geq 8$.

In the odd case, identify the point labels with $\mathbb{Z}/n\mathbb{Z}$ by subtracting one. Then x is translation by one. Any permutation inverting x has the form $z \mapsto c - z$: its value at zero determines all its other values by the inversion equation. The two transpositions of t have supports $\{0, 1\}$ and $\{2, 5\}$. A reflection preserves their cyclic differences up to sign. Since $1 \not\equiv \pm 3 \pmod{n}$ for $n \geq 9$, it cannot interchange these two supports. Preserving them separately would require simultaneously $c \equiv 1$ and $c \equiv 7 \pmod{n}$, which is impossible. This completes both chirality arguments. □

The usual correspondence between chiral polyhedra and generating pairs (x, t), with t an involution and no inverting automorphism fixing t, therefore supplies the alternating-group existence branch of the source theorem. That correspondence is a retained input, attributed to Schulte–Weiss as in the source. The theorem above proves the actual group conditions. It does not assert the other classical or exceptional Lie families, sporadic cases, their computer searches, or the almost-simple question in the concluding section. Those obligations remain separate.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, fixed retained arXiv:1603.07713v3 PDF, arXiv header 18 May 2016, internal date 6 November 2018, Section 5. The displayed generating pairs are attributed there to Gareth Jones's February 2016 Queenstown lecture.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991),493–516.

Chirality detected by the order of one word

Human Compression project

We give finite certificates for twelve sporadic cases of the existing chiral-polyhedron existence theorem of Leemans and Liebeck [1]. The same short word detects chirality in every case. The two Mathieu pairs printed in that paper are retained verbatim; the other pairs are explicit new witnesses to already known existence assertions. Six of their ambient representations are acquired from our earlier, separately reviewed certificates. None of these cases is claimed as a new existence theorem or as a solution to an open conjecture.

1 The common criterion

If w is a word in two letters, write $w(x,t)$ for its value at x,t.

Proposition 1. *Suppose $G=\langle x,t\rangle$ is a finite group, t has order two, and some word w satisfies*

$$|w(x,t)|\neq|w(x^{-1},t)|.$$

No automorphism of G can invert x and fix t.

Proof. An automorphism with this action would send $w(x,t)$ to $w(x^{-1},t)$. An isomorphism preserves element orders, contradicting the displayed inequality. This excludes all automorphisms with the stated action, including the involutory ones in the geometric criterion. □

The proposition is an elementary use of word invariants, not a new automorphism theorem. It is useful because a single checked inequality replaces a search through the automorphism group.

For completeness, suppose also that $Z(G)=1$, and put $y=x^{-1}t$. An element of $\langle x\rangle\cap\langle y\rangle$ commutes with both generators x,y, and hence belongs to $Z(G)$. Thus

$$\langle x\rangle\cap\langle y\rangle=1,\qquad (xy)^2=1.$$

In all the examples below $|x|,|y|>2$. These are precisely the rank-three rotation intersection and product conditions. We retain the geometric reconstruction theorem of Schulte and Weiss [2]: these data give a directly regular or chiral polyhedron, and the automorphism in Proposition 1 is the regularity alternative. Its absence gives a chiral polyhedron of type $\{|x|,|y|\}$ with full automorphism group G. The sporadic groups below are nonabelian simple groups, so their centres are trivial. Their identification and simplicity are explicit retained classification inputs.

2 Twelve exact instances

In every row use the single word

$$w(x,t)=xtx^2tx^3t.$$

The last two columns differ in every case.

G	$\lvert G\rvert$	$\lvert x\rvert$	$\lvert x^{-1}t\rvert$	$\lvert w(x,t)\rvert$	$\lvert w(x^{-1},t)\rvert$
M_{11}	7920	11	8	8	3
M_{12}	95040	8	8	10	11
M_{22}	443520	8	7	7	5
M_{23}	10200960	11	15	14	6
M_{24}	244823040	23	15	15	12
J_1	175560	10	6	6	19
J_2	604800	15	15	10	15
J_3	50232960	10	9	4	17
HS	44352000	8	7	7	11
He	4030387200	14	15	17	21
Suz	448345497600	10	9	12	8
McL	898128000	30	8	14	11

These rows assert existence. They do not recover the source's ancillary counts of nonisomorphic polyhedra or require its particular example type when we supply a different witness.

The files `mathieu_pairs.g`, `rank4_pairs.g` and `small_pairs.g` specify every permutation. For M_{23}, M_{24}, the source gives x, y and we set $t = xy$. The checker tests the two permutations literally and computes their orders. For M_{23} it checks their actual membership in the named Mathieu group. For M_{24} the displayed labelling differs from the installed standard one, so the checker constructs a bijective homomorphism onto that named group. Equality of orders alone is not used to identify it.

For $M_{12}, J_2, HS, He, Suz, J_3$, let (r_1, r_2, r_3, r_4) be the acquired four-involution marking. The saved list of thirty-two indices $(i_1, \ldots, i_{32})$ defines

$$x = r_{i_1} \cdots r_{i_{32}}, \qquad t = r_1.$$

The whole marking, its ambient generators and the named-group identification are bound inputs. For the first five groups the ambient extensions are the following primitive catalogue entries:

Simple socle	degree	catalogue index
M_{12}	144	4
J_2	100	2
HS	100	4
He	2058	2
Suz	1782	2

Their derived subgroups are the specified simple groups. The $J_3 : 2$ representation and defining presentation are acquired from the earlier certificate and its Atlas inputs [3]. We retain those named identifications; we do not charge them as if the group name followed from an order.

The remaining four pairs are literal permutations in `MathieuGroup(11)`, `MathieuGroup(22)`, the derived subgroup of `PrimitiveGroup(275,2)`, and `PrimitiveGroup(266,1)`. The last catalogue record explicitly identifies J_1; that identification is a retained input too. The checker tests membership of both permutations in each named group before comparing the generated subgroup's order with its order.

3 Certificate and scope

For every saved pair, the finite checker verifies $t^2 = 1$, $t \neq 1$, membership in the specified group, generation of the full group, and each of the four orders in the table. Since the group is finite, a contained subgroup of equal order is the whole group. This proves generation from the saved

data. Proposition 1 then proves the absence of reversal. The geometric conclusion uses only the retained reconstruction theorem and the already identified simple group.

The witness searches are construction aids and are not replay inputs. No search over candidate pairs or automorphisms occurs in verification. GAP's exact permutation and subgroup algorithms are computational proof infrastructure; they have not been replaced here by Lean proofs. The complete checker, saved data and all required group identifications remain inside the dependency boundary for any later comparison.

The other fourteen sporadic cases and the classical and exceptional Lie families in the source remain separate obligations. This component does not complete the original manuscript. The existing existence results, the two printed Mathieu witnesses, the external group classification and the geometric reconstruction theorem retain their attribution; the new saved witnesses and this common checking argument are the contribution of this component.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, retained arXiv:1603.07713v3, Sections 1 and 6.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516.

[3] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[4] Human Compression project, *Explicit sporadic markings and their incidence certificates*, accepted N006 component; exact files and admission recorded in the dependency ledger.

Counting involutions that generate a chiral polyhedron

Human Compression project

We give one counting argument for fourteen sporadic cases of the existing existence theorem of Leemans and Liebeck [1]. It uses ordinary class sizes and a complete list of maximal subgroup classes. Both are retained classification inputs, with their exact numerical use recorded. The same involution bound handles all fourteen cases without distinguishing the conjugacy classes of reversing automorphisms.

1 The common counting theorem

For a finite group H, let $i_d(H)$ be the number of elements of exact order d. Let G be nonabelian simple and identify G with its inner automorphism group in $A = \mathrm{Aut}(G)$. Fix $x \in G$ of odd prime order p. Choose one representative of each G-conjugacy class of maximal subgroups, and define

$$B_G(x) = |C_G(x)| \sum_M \frac{i_p(M) i_2(M)}{|M|}. \tag{1}$$

Let μ be an upper bound for $i_2(C_G(\alpha))$ for every involution $\alpha \in A$ that inverts x. Put

$$R_A(x) = \begin{cases} 0, & x \text{ is not conjugate to } x^{-1} \text{ in } A, \\ |C_A(x)|\mu, & \text{otherwise.} \end{cases} \tag{2}$$

The upper bound in (1) may be rational; it need not count an actual set exactly.

Theorem 1. *If $i_2(G) > B_G(x) + R_A(x)$, there is an involution $t \in G$ such that $G = \langle x, t \rangle$ and no automorphism of G inverts x while fixing t.*

Proof. Every maximal subgroup $M < G$ is self-normalizing. Indeed its normalizer is M or G, and the latter would make M a proper nontrivial normal subgroup of the simple group G. Count the conjugates of M containing x. The set of conjugating elements taking x into M has size $|C_G(x)|\,|x^G \cap M|$. Each conjugate of M is counted $|M|$ times. Hence the number of these conjugates is

$$\frac{|C_G(x)|\,|x^G \cap M|}{|M|} \leq \frac{|C_G(x)| i_p(M)}{|M|}.$$

If $\langle x, t \rangle$ is proper, it is contained in one of these maximal subgroups. Summing their involution counts proves that at most $B_G(x)$ involutions fail to generate with x.

If an automorphism α inverts x and fixes a generating partner t, then α^2 fixes both generators and is the identity. Since p is odd, α is nonidentity and thus an involution. All elements of A taking x to x^{-1}, if any exist, form a coset of $C_A(x)$: the quotient of any two such elements centralizes x. There are at most $|C_A(x)|$ possible inverting involutions. Each fixes at most μ involutions of G. The union of these fixed sets has size at most $R_A(x)$; it is empty if x is not conjugate to its inverse in A. The strict inequality therefore leaves an involution outside both excluded sets. It has the required properties. □

The maximal-subgroup count and inverter coset are standard counting arguments, closely related to the preliminary lemmas of [1]. Our common specialization uses a deliberately coarse bound for μ. It avoids a separate analysis of the reversing involutions in each group.

Lemma 2. *If $A = G$, one may take*

$$\mu = \max_{M \text{ maximal in } G} i_2(M). \tag{3}$$

Suppose instead that $[A : G] = 2$ and $|C_A(a)| = 2|C_G(a)|$ for every involution $a \in G$. Then one may take

$$\mu = \max_{M \text{ maximal in } A,\ M \neq G} i_2(M). \tag{4}$$

Proof. The centre of A is trivial. An automorphism central in A commutes with every inner automorphism; therefore it fixes every element of the centreless group G and is the identity. In particular the centralizer of a nonidentity involution in A is proper and lies in a maximal subgroup. This proves (3), since $C_G(\alpha) \leq C_A(\alpha)$.

For (4), if $\alpha \notin G$, its centralizer contains α and is not contained in G. If $\alpha \in G$, the stated centralizer index likewise shows that $C_A(\alpha)$ contains an element outside G. A maximal subgroup containing this centralizer therefore cannot be G. Containment bounds its involution count by (4), and again $C_G(\alpha) \leq C_A(\alpha)$. □

The centralizer condition in the second part is checked for every involution class of G, using its actual fusion into A. It is not inferred merely from the fact that A/G has order two.

2 Fourteen applications

For each group below choose the specified prime-order class in the bound data. The column L is a lower bound for $i_2(G)$, and B, R are the exact upper bounds computed from (1)–(4). In every row $B + R < L$. Only for B and M do we replace the exact involution count by a smaller power of ten in this display; the full counts remain in the certificate.

G	p	L	B	R
J_4	29	51747149311	29	101486051
Fi_{23}	23	12895196031	976120	0
Fi'_{24}	29	7824165773823	58	747921369827
M	71	10^{20}	4970	0
B	47	10^{12}	0	0
Th	31	976841775	44826	0
Ly	37	1296826875	74	15054375
Co_1	23	12792894975	72100538	0
Co_2	23	29771775	7590	0
Co_3	23	2778975	7590	0
Ru	29	1846575	870	390195
Fi_{22}	13	37706175	1443546	18669235
$\mathrm{O'N}$	31	2857239	1860	95945
HN	19	75603375	7182	2662869

The full automorphism groups equal G except in the four rows $\mathrm{Fi}'_{24}, \mathrm{Fi}_{22}, \mathrm{O'N}, HN$, where they are the retained extensions of index two. The ordinary tables and their actual class fusions verify the condition in Lemma 2 in each of those four cases. For the rows with $R = 0$, the inverse class differs from the chosen class in the full automorphism group, not merely in an unspecified subgroup.

For clarity, the nonzero values of μ actually used are

G	μ	G	μ
J_4	3499519	Fi_{22}	1436095
Ly	406875	$\mathrm{O'N}$	3095
Fi'_{24}	25790392063	HN	140151
Ru	13455		

These are maxima over complete maximal-subgroup lists, as specified in (3) or (4). For example, the largest value for the automorphism group of Fi'_{24}, after omitting its simple socle, is $i_2(2 \times \mathrm{Fi}_{23}) = 25790392063$. Thus even this coarse bound suffices. No assertion that an inverter belongs to a particular class of involutions is needed.

Theorem 1 gives the required generating pairs. For the polyhedron conclusion put $y = x^{-1}t$. The cyclic intersection $\langle x \rangle \cap \langle y \rangle$ is central in $\langle x, y \rangle = G$, so it is trivial. Also $(xy)^2 = 1$. The element y cannot be an involution: otherwise G would be generated by the two involutions y, t and would be dihedral. The case $y = 1$ would make G cyclic. Hence $|x|, |y| > 2$. The retained Schulte–Weiss reconstruction theorem supplies a chiral polyhedron with full automorphism group G, since the reversing automorphism has been excluded.

3 Inputs, verification, and attribution

The source's existing existence conclusions are being reproved, not claimed as new. The retained facts are the identifications of the sporadic groups and their automorphism groups, the complete lists of maximal subgroup classes, the ordinary class sizes and element orders, and the displayed inclusion fusions. The verifier obtains the numerical data from GAP's Character Table Library, in its recorded edition. Its `Maxes` attribute is documented to give one representative for every conjugacy class of maximal subgroups when it is present. Completeness of those classification lists is a premise; it is not proved by iterating over a software list.

For every required table, the checker sums the sizes of the classes of exact order two or p. It retains every maximal-subgroup class, including repetitions of a table name when distinct subgroup classes have the same character table. It verifies each centralizer index used in (4), the actual inverse-class test in (2), and all fourteen strict inequalities with exact integers or rationals. The full underlying rows and their source bindings are retained, so the small displayed table is not an unexplained set of trusted totals.

Together with the separately reviewed twelve explicit word certificates, these fourteen cases cover all twenty-six sporadic groups in the source theorem. The alternating branch has its own uniform proof. The classical and exceptional Lie families remain separate obligations.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, fixed arXiv:1603.07713v3, Sections 2 and 6.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991),493–516.

[3] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[4] T. Breuer, *The GAP Character Table Library*, version 1.3.11, data and manual Section 3.7.1; the exact retained installation files are specified in the dependency ledger.

Finite Lie-group witnesses for chiral polyhedra

Human Compression project

1 The finite source obligations

Leemans and Liebeck [1] prove that every nonabelian finite simple group except $A_7, \mathrm{PSL}_2(q), \mathrm{PSL}_3(q)$ and $\mathrm{PSU}_3(q)$ admits a chiral polyhedron. We recover the existence assertions for the thirty-five parameter entries in Table 1. These are existing results, not new existence theorems.

There are twenty-one simple entries in their list (4.1), physical page 7:

$$L_4^\epsilon(2), L_4^\epsilon(3), L_4^\epsilon(4), L_5^\epsilon(2), L_6^\epsilon(2),$$
$$\mathrm{PSp}_4(3), \mathrm{Sp}_4(4), \mathrm{Sp}_6(2), \mathrm{PSp}_6(3), \mathrm{Sp}_8(2),$$
$$\Omega_7(3), \Omega_8^\epsilon(2), \mathrm{P}\,\Omega_8^+(3), \Omega_{10}^\epsilon(2).$$

Both signs occur, with $L^+ = \mathrm{PSL}$ and $L^- = \mathrm{PSU}$. The additional printed $\mathrm{Sp}_4(2) \cong S_6$ is not simple; it is not a positive simple-case obligation. The eleven remaining finite exceptions in Section 4(2), physical page 8, are

$$L_5^\epsilon(3), L_6^\epsilon(3), L_8^\epsilon(2), \mathrm{Sp}_6(4), \mathrm{PSp}_8(3), \mathrm{Sp}_8(4), \Omega_{12}^\epsilon(2).$$

The last three entries are $G_2(3), G_2(4), {}^2F_4(2)'$, treated at the end of Section 3, physical page 6. We do not include the nonsimple $G_2(2)$, nor turn its derived group $\mathrm{PSU}_3(3)$ or ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$ into positive exceptions.

Lemma 1. *Let $G = \langle x, t\rangle$, where $|t| = 2$. If $|w(x,t)| \neq |w(x^{-1},t)|$ for a group word w, no automorphism of G inverts x and fixes t.*

Proof. Such an automorphism would carry the first word value to the second. It preserves element order, which contradicts the inequality. □

This is the elementary word criterion already proved in the accepted twelve-sporadic component [4]; it is restated to make the present argument self-contained. In every row we take $w(x,t) = xtx^2tx^3t$.

Theorem 1. *Each parameter entry in Table 1 has generators x, t satisfying conditions (i)–(iii) of [1, Theorem 1.1]. It therefore admits a chiral polyhedron of the displayed type $\{|x|, |x^{-1}t|\}$, with that named simple group as full automorphism group.*

Proof. The exact checks described below establish membership, $|t| = 2$, full generation, and the unequal word orders. Lemma 1 rules out even noninvolutory reversing automorphisms. Put $y = x^{-1}t$. Then $xy = t$. An element of $\langle x\rangle \cap \langle y\rangle$ commutes with both x, y, so is central. The named groups are nonabelian simple, hence this intersection is trivial. The first two orders exceed two. The retained Schulte–Weiss reconstruction theorem [2], in the precise form used in [1, Section 1], now gives the stated geometric conclusion. □

2 Named models and exact verification

Every `pair.g` is a literal permutation record containing the two witnesses, the full ambient generating list, the parameters, the degree and the expected orders. A fresh GAP process resets the same recorded constructor seed and calls `SimpleGroup` with those parameters. It requires exact equality of the saved and newly constructed generating lists. There is no recognition of a group from its order.

The named-model identifications are retained GAP 4.15.1 construction and catalogue premises [3]. The `L`, `U` and `S` branches construct `PSL`, `PSU` and `PSp`. Their projective matrix constructions and subsequent faithful smaller actions are the documented named constructors. The odd and signed even orthogonal branches take the derived subgroup of the appropriate special orthogonal action on lines, followed when applicable by the constructor's smaller action. Their identification and simplicity are retained; neither the stored `Size` nor a simplicity flag proves the identification here. The three exceptional calls are explicitly

$G_2(3)$	`PrimitiveGroup(351,7)`
$G_2(4)$	`PrimitiveGroup(416,7)`
${}^2F_4(2)'$	`PrimitiveGroup(1600,20).`

The exact primitive catalogue records name these groups. The constructor, classical projective interface, primitive loader and actual catalogue files are pinned in the dependency ledger. Correctness of GAP's exact permutation algorithms and the named library inputs is explicit computational infrastructure, not a new classification proof.

For every row the checker reconstructs H from the saved ambient generators without supplying its order. It verifies $H = G$, computes $|H|$, checks $x, t \in H$ and $|t| = 2$, and forms $K = \langle x, t \rangle$ without supplying its order. It verifies $|K| = |H|$ and $K = H$. Membership and equal finite order already imply generation; the latter equality is a redundant explicit check. The four table entries are then computed by exact permutation operations. The new groups are checked to have no preassigned size before the relevant calculations.

A separate standard-library Python parser reads the original cycle notation directly, checks disjoint cycles, and composes the corresponding right-action permutations. Cycle lengths and integer least common multiples reproduce all four orders and $|t| = 2$, independently of GAP's word calculations. It also parses every ambient generator, but it does not purport to duplicate GAP's large subgroup computations.

The source roster is constructed independently as the explicit sets of parameters above. Missing, duplicate, nonsimple, misrouted and displaced entries are rejected. Seven GAP controls damage parameters, ambient generators, the involution, membership, generation, an order, and the word inequality. All must reject before producing a positive certificate. Both Python entry points reject optimized execution before output work. Every replay is finite; a timeout or nonclean terminal result grants no certificate. The recorded searches merely supplied candidates and are not run by verification.

3 Attribution and limits

The root agent supplied the bounded candidate searches and the acquired word argument; this component's author supplies the deterministic certificate, independent literal oracle, source routing and explanation. All original candidate and execution bytes are retained. An initial author setup used a directory ordinal instead of the actual recorded seed for some constructors; that failed preliminary setup is preserved outside the scientific replay. The final replay uses the recorded seed and changes no candidate.

These are thirty-five parameter entries, not thirty-five newly classified groups. In particular $L_4(2) \cong A_8$ overlaps the alternating family, and $U_4(2) \cong \mathrm{PSp}_4(3)$ occurs twice under the source's parameters. The group identities, geometric reconstruction and original existence theorem retain their attribution. This component does not settle the infinite Lie families, ancillary enumerations, or the whole original source, and adds no measurement or conjecture claim.

4 The complete finite table

G	$\lvert x\rvert$	$\lvert x^{-1}t\rvert$	$\lvert w\rvert$	$\lvert w^{-}\rvert$
$L_4(3)$	8	5	9	12
$U_4(3)$	8	9	7	5
$L_4(4)$	30	85	30	15
$U_4(4)$	65	30	65	15
$L_5(2)$	15	15	3	7
$U_5(2)$	12	18	12	11
$L_4(2)$	15	6	5	6
$U_4(2)$	9	12	6	5
$L_6(2)$	12	31	28	31
$U_6(2)$	6	18	10	11
$S_4(3)$	4	9	4	5
$S_4(4)$	15	5	17	10
$S_6(2)$	6	9	12	10
$S_6(3)$	13	9	24	9
$S_8(2)$	30	30	8	6
$O_7(3)$	13	13	7	8
$O_8^+(2)$	12	10	10	12
$O_8^-(2)$	30	6	7	6
$O_8^+(3)$	15	12	15	20
$O_{10}^+(2)$	12	31	12	15
$O_{10}^-(2)$	11	30	6	15
$L_5(3)$	121	9	18	121
$U_5(3)$	24	28	21	40
$L_6(3)$	52	121	10	39
$U_6(3)$	91	80	13	56
$L_8(2)$	30	105	62	124
$U_8(2)$	85	66	12	85
$S_6(4)$	51	63	20	63
$S_8(3)$	13	84	24	40
$S_8(4)$	195	126	255	65
$O_{12}^+(2)$	63	18	24	18
$O_{12}^-(2)$	24	102	34	60
$G_2(3)$	13	12	7	6
$G_2(4)$	7	13	6	15
${}^2F_4(2)'$	12	8	13	12

Table 1: Exact orders in the saved witnesses. $L_n(q), U_n(q), S_n(q)$ mean $\mathrm{PSL}_n(q), \mathrm{PSU}_n(q), \mathrm{PSp}_n(q)$; $O_n^{\pm}(q)$ means $\mathrm{P}\,\Omega_n^{\pm}(q)$ and $O_7(3) = \Omega_7(3)$. Let $w = xtx^2tx^3t$ and $w^- = w(x^{-1}, t)$.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, fixed arXiv:1603.07713v3, Sections 1, 3 and 4.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516.

[3] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1, reference manual Sections 39.15-14 and 50.2; PrimGrp 4.0.1. Exact local implementation and data pins accompany this proof.

[4] Human Compression project, *Chirality detected by the order of one word*, accepted twelve-sporadic component, final edition v2; its distinct review and root adoption are retained.

Explicit involution classes for the classical part of L045

Private proof component: publication_recovery

This note proves the lower estimates needed in Section 4 of the fixed L045 source, and corrects its special unitary degree-four estimate. It does not establish the required maximal-overgroup or automorphism bounds, or assert a complete recovery of the original paper. Throughout, $\mathrm{i}_2(H)$ counts elements of order exactly two. The finite 35 parameter certificates are a separate, already distinctly reviewed component.

Theorem 1. *For every prime power q, the following strict lower bounds hold:*

G	*range*	$\mathrm{i}_2(G) > q^D/4$
$\mathrm{PSL}_n(q),\ \mathrm{PSU}_n(q)$	$n \geq 4$	$D = \lfloor n^2/2 \rfloor$
$\mathrm{PSp}_{2m}(q)$	$m \geq 2$	$D = m^2 + m$
$\Omega_{2m+1}(q)$	$m \geq 3,\ q$ *odd*	$D = m^2 + m$
$\mathrm{P}\,\Omega^{\epsilon}_{2m}(q)$	$m \geq 4,\ \epsilon = \pm 1$	$D = m^2 - 1.$

The nonsimple group $\mathrm{Sp}_4(2)$ is allowed in this numerical statement; it is not a positive simple-group obligation in L045.

Classical conventions and elementary counts. We use the natural finite-field matrix groups. In odd characteristic, Ω is the kernel of the spinor norm in the determinant-one orthogonal group; in even characteristic it is the kernel of the Dickson invariant $g \mapsto \mathrm{rank}(g - I) \bmod 2$. These are the usual matrix descriptions of the groups denoted Ω in the source. A projective group means the quotient by the scalar centre. The ordinary finite-field form conventions, including the uniqueness of a nondegenerate hermitian form in each dimension and the two signs of an even-dimensional quadratic form, are retained elementary classical-group inputs. No maximal-subgroup or involution-class classification is used below.

Put $P_j(q) = \prod_{i=1}^{j}(1 - q^{-2i})$ and $F_j(q) = \prod_{i=1}^{j}(1 - (-q)^{-i})$, with empty products one. Counting ordered bases, symplectic bases, orthogonal bases, and unitary bases gives

$$|\,\mathrm{GL}_j(q)| = q^{j^2}\prod_{i=1}^{j}(1 - q^{-i}), \qquad |\,\mathrm{GU}_j(q)| = q^{j^2}F_j(q),$$

$$|\,\mathrm{Sp}_{2j}(q)| = q^{2j^2+j}P_j(q), \qquad |O^{\epsilon}_{2j}(q)| = 2q^{2j^2-j}(1 - \epsilon q^{-j})P_{j-1}(q),$$

$$|O_{2j+1}(q)| = 2q^{2j^2+j}P_j(q) \qquad (q \text{ odd}).$$

We count whole ambient conjugacy orbits of matrices lying in the desired special group. They need not be single conjugacy classes of that special group. If X^2 is scalar, at most two scalar multiples of X can lie in its ambient class: $(zX)^2 = X^2$ forces $z^2 = 1$. If the two eigenvalue multiplicities are unequal, only $z = 1$ is possible. Thus the divisions by two below are valid lower bounds even when special-group classes split.

Lemma 2. *For $q \geq 2$, $F_j(q) \leq 1 + q^{-1}$. A consecutive interval of its factors starting at a has product at least $1 - q^{-a}$, and has product greater than one if a is odd. Also*

$$\prod_{i \geq a}(1 - q^{-2i}) > 1 - \frac{q^{-2a}}{1 - q^{-2}}, \qquad \prod_{\substack{i \geq 1 \\ i\ odd}}(1 - q^{-i}) > 1 - \frac{q}{q^2 - 1}.$$

In the last inequality we use $q \geq 3$; the finite products used below obey the same strict estimates.

Proof. Pair an even factor $1 - q^{-2i}$ with its following odd factor to obtain a product less than one; this proves the upper bound for F_j. Pair an odd factor with its following even factor to obtain a product greater than one; an interval beginning at an even index loses only its first factor. The remaining inequalities follow by expanding a product of numbers $1 - a_i$, $0 < a_i < 1$, and bounding it below by $1 - \sum a_i$. □

Linear and unitary groups in odd characteristic. For odd n choose an even k nearest $n/2$ and take the matrix with -1-eigenspace of dimension k. For even $n = 2m$ with even m, take the balanced $m + m$ decomposition. All these matrices have determinant one; use orthogonal hermitian summands in the unitary case. Their ambient orbits have sizes $|\operatorname{GL}_n|/(|\operatorname{GL}_k||\operatorname{GL}_{n-k}|)$ or $|\operatorname{GU}_n|/(|\operatorname{GU}_k||\operatorname{GU}_{n-k}|)$, respectively. The normalized linear ratio exceeds one. For the unitary ratio, $k, n-k \geq 2$, so Lemma 2 gives

$$\frac{F_n(q)}{F_k(q)F_{n-k}(q)} > \frac{1-q^{-3}}{1+q^{-1}} \geq \frac{13}{18}.$$

The orbit exponent is $2k(n-k) = \lfloor n^2/2 \rfloor$. Division by at most two still leaves a constant exceeding $1/4$.

If $n = 2m$ and m is odd, use a determinant-one matrix X with $X^2 = -I$. In the linear case with $q \equiv 1 \pmod 4$, take $\operatorname{diag}(aI_m, -aI_m)$, $a^2 = -1$; with $q \equiv 3 \pmod 4$, take multiplication by a on $\mathbb{F}_{q^2}^m$. The centralizers are $\operatorname{GL}_m(q)^2$ and $\operatorname{GL}_m(q^2)$. In the unitary case the two congruences are reversed: a has norm one if $q \equiv 3 \pmod 4$, while for $q \equiv 1 \pmod 4$ the two eigenspaces are paired totally isotropic spaces, and the centralizer is $\operatorname{GL}_m(q^2)$. In the latter cases the respective normalized orbit ratios are

$$\prod_{\substack{1 \leq i \leq 2m-1 \\ i \text{ odd}}} (1-q^{-i}) > 5/8, \qquad \prod_{\substack{1 \leq i \leq 2m-1 \\ i \text{ odd}}} (1+q^{-i}) > 1.$$

The exponent is $2m^2$, and projectivization loses at most two.

Linear and unitary groups in characteristic two. Write $n = 2k+l$, $l \in \{0,1\}$. A linear involution of Jordan shape $(2^k, 1^l)$ has centralizer $q^{k(2n-3k)} \operatorname{GL}_k(q) \operatorname{GL}_l(q)$. This is obtained by writing commuting endomorphisms along its image/kernel filtration: the invertible diagonal blocks are one repeated k-block and one l-block. Its normalized orbit ratio is

$$\frac{\prod_{i=k+1}^{n}(1-q^{-i})}{\prod_{i=1}^{l}(1-q^{-i})} > 3/4.$$

For a hermitian space, put $X = I + N$. Unitarity and $N^2 = 0$ imply that $W = \operatorname{im} N$ is totally isotropic and N is self-adjoint. Conversely, for each isotropic k-space W, the rank-k self-adjoint maps $V/W^\perp \longrightarrow W$ are precisely the nonsingular hermitian matrices; their number is $|\operatorname{GL}_k(q^2)|/|\operatorname{GU}_k(q)|$. The stabilizer of W has order $q^{k(2n-3k)}|\operatorname{GL}_k(q^2)||\operatorname{GU}_l(q)|$: the radical has $2kl + k^2$ parameters over $\mathbb{F}_q$, and the indicated two diagonal blocks are free. Therefore the total number is

$$\frac{|\operatorname{GU}_n(q)|}{q^{k(2n-3k)}|\operatorname{GU}_k(q)||\operatorname{GU}_l(q)|}.$$

Its exponent is $2k(n-k)$ and its normalized ratio $F_n/(F_kF_l)$ exceeds $(1-2^{-3})/(1+2^{-1}) = 7/12$. All matrices just counted are unipotent, hence determinant one. In characteristic two no nontrivial scalar multiple of a unipotent matrix is unipotent. This proves both projective lower bounds.

Symplectic groups in odd characteristic. Use a symplectic complex structure $X^2 = -I$. If $q \equiv 1 \pmod 4$, its two isotropic eigenspaces are paired and its centralizer is $\operatorname{GL}_m(q)$. If $q \equiv 3 \pmod 4$, view the space as a hermitian m-space over $\mathbb{F}_{q^2}$ and obtain centralizer $\operatorname{GU}_m(q)$. These statements follow directly by requiring a commuting map to preserve the pairing or the hermitian form, respectively. The orbit exponent is $m^2 + m$; the normalized ratios are

$$\prod_{i=1}^{m}(1+q^{-i}) > 1, \qquad \prod_{i=1}^{m}(1+(-q)^{-i}) > 5/8.$$

At most two-to-one scalar fusion gives the result.

Symplectic groups in characteristic two. Fix an isotropic k-space W. The symplectic rank-k square-zero maps with image W correspond to nonsingular symmetric $k \times k$ matrices. Retain just the nonalternating ones. Every nonalternating nondegenerate symmetric form over the perfect field $\mathbb{F}_q$ is equivalent to I_k. One may diagonalize after splitting a nonsingular vector; a remaining alternating plane together with one nonsingular line is equivalent to three nonsingular lines. Its isometry group has order

$$J_k(q) = \begin{cases} |\operatorname{Sp}_{k-1}(q)|, & k \text{ odd}, \\ q^{k-1}|\operatorname{Sp}_{k-2}(q)|, & k \text{ even}. \end{cases}$$

Indeed the distinguished vector representing $v \mapsto \sqrt{(v,v)}$ is fixed. If it is nonsingular its perpendicular complement is symplectic; if it is singular the quotient of its perpendicular space is symplectic and the kernel, by the two remaining row/column equations, has q^{k-1} elements. The isotropic-space stabilizer has Levi part $\mathrm{GL}_k(q)\,\mathrm{Sp}_{2m-2k}(q)$ and radical order q^u, where $u = (4mk - 3k^2 + k)/2$ (count the free off-diagonal block and its symmetric correction). Consequently the number of these involutions is exactly

$$C_{m,k}(q) = \frac{|\,\mathrm{Sp}_{2m}(q)|}{q^u|\,\mathrm{Sp}_{2m-2k}(q)|J_k(q)}. \tag{1}$$

For $k = m$ this is q^{m^2+m} times $\prod_{i=\lceil m/2\rceil}^{m}(1 - q^{-2i}) > 2/3$. The symplectic centre is trivial in characteristic two.

Orthogonal groups in odd characteristic. For odd dimension $2m + 1$, let $2a$ be the even member of $\{m, m + 1\}$, and $b = m - a$. Choose a nondegenerate $2a$-space with square discriminant; such a space of one of the two signs exists with a nondegenerate complement. Acting as $-I$ on it and I on its complement has determinant and spinor norm one. The full orthogonal orbit has size

$$\frac{|O_{2m+1}(q)|}{|O^{\delta}_{2a}(q)||O_{2b+1}(q)|}.$$

All its elements lie in Ω. The exponent is $m^2 + m$, and its normalized ratio exceeds $(1/2)(7/8)/(4/3) = 21/64$.

For dimension $2m$, put $a = \lfloor m/2\rfloor$, $b = m - a$, choose again a square-discriminant $2a$-space, and use the same sign involution. Its orthogonal orbit has size

$$\frac{|O^{\epsilon}_{2m}(q)|}{|O^{\delta}_{2a}(q)||O^{\epsilon\delta}_{2b}(q)|}.$$

The exponent is $4ab$. Apart from the leading factor $1/2$, its normalized ratio is greater than

$$\frac{(1 - 3^{-4})(7/8)}{(1 + 3^{-2})^2} = 7/10.$$

If m is odd the two multiplicities differ, so the projective map is injective on this orbit and $4ab = m^2 - 1$. If m is even it loses at most two and $4ab = m^2$; the extra factor $q \geq 3$ more than pays this loss.

Orthogonal groups in characteristic two. Use (1) with even $k = m$ for even m, and $k = m - 1$ for odd m. In a hyperbolic symplectic basis the chosen involution sends f_i to $f_i + e_i$ for $1 \leq i \leq k$ and fixes all other basis vectors. A quadratic form Q with this polar form is invariant precisely when $Q(e_i) = 1$ for these k indices. Thus exactly q^{2m-k} forms are invariant. Their Arf trace is $\mathrm{Tr}(\sum Q(e_i)Q(f_i))$; since $k > 0$, one free $Q(f_i)$ makes its two values equally frequent. Exactly $q^{2m-k}/2$ invariant forms have each sign. There are $q^m(q^m + \epsilon)/2$ forms of sign ϵ with the fixed polar form. Double counting pairs (X, Q) gives

$$C_{m,k}(q)\frac{q^{m-k}}{q^m + \epsilon} \tag{2}$$

involutions in each orthogonal group. Their Dickson invariant is $k \bmod 2 = 0$, so all belong to Ω; its scalar centre is trivial here. For even m, (2) is

$$q^{m^2}\frac{\prod_{i=m/2}^{m}(1 - q^{-2i})}{1 + \epsilon q^{-m}}.$$

For odd m it is

$$q^{m^2-1}\frac{\prod_{i=(m-1)/2}^{m}(1 - q^{-2i})}{(1 - q^{-2})(1 + \epsilon q^{-m})}.$$

Both normalized ratios exceed $(11/12)/(17/16) = 44/51$. This finishes Theorem 1.

Proposition 3. *If $q \equiv 3 \pmod 4$, then*

$$\mathrm{i}_2(\mathrm{PSU}_4(q)) = \frac{q^4(q^2 - q + 1)(q^2 + 1)}{2} + \begin{cases} 0, & q \equiv 3 \pmod 8, \\ q^3(q - 1)(q^2 + 1), & q \equiv 7 \pmod 8. \end{cases}$$

In particular the source's special assertion $\mathrm{i}_2(\mathrm{PSU}_4(q)) > q^8/2$ is false for all such q.

Proof. A projective involution lifts to $X \in \mathrm{SU}_4(q)$ with $X^2 = zI$, $z \in \mu_4$. It is semisimple. For $z = 1, -1$, determinant one forces the noncentral multiplicities to be $2+2$. Both choices give the same projective family; its size is $|\,\mathrm{GU}_4\,|/(2|\,\mathrm{GU}_2\,|^2)$, the first summand displayed above. For z of order four, choose $a^2 = z$, so a has order eight. If $q \equiv 3 \pmod 8$, $a^{q+1} = -1$; its two eigenspaces must be paired isotropic spaces of dimensions $2+2$. But the determinant is then $a^4 = -1$, impossible. If $q \equiv 7 \pmod 8$, $a^{q+1} = 1$; the eigenspaces are nondegenerate and determinant one requires multiplicities $1+3$ or $3+1$. Scalar multiplication cycles all four choices of a and identifies the two multiplicity orientations, leaving one projective family. Within one $1+3$ ambient orbit the projective map is injective, so its size is $|\,\mathrm{GU}_4\,|/(|\,\mathrm{GU}_1\,||\,\mathrm{GU}_3\,|) = q^3(q-1)(q^2+1)$. There are no remaining possibilities for z or for the eigenspaces.

For $q \equiv 3 \pmod 8$, subtracting $q^8/2$ from the first summand gives $-(q^4/2)(q(q-1)^2-1) < 0$. For $q \equiv 7 \pmod 8$, the difference is $-(q^3/2)(q^4-4q^3+3q^2-3q+2) < 0$, since $q \geq 7$. □

Exact controls and attribution. The accompanying standard-library program evaluates all written class formulas with rational arithmetic at $1,619$ parameter choices, verifies the strict quarter bounds, and checks the correction at seven fields. These finite checks are controls on the formulas, not premises for the uniform inequalities. The ordinary character table $\mathrm{U}_4(3)$ gives $2,835$ involutions, agreeing with Proposition 3; at $q = 7$ the formula gives $2,683,975 < 7^8/2$. A fresh clean table replay is retained with the component. The construction and calculations in this note are new recovery work by publication_recovery, responding to root's request for explicit lower estimates. The fixed source is Leemans–Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, Lemma 2.4 and Section 4 (especially printed page 9). No claim of novelty or priority is made for the classical formulas. The remaining existence argument must separately pay every bad-generator and reverser bound.

Torus reversers and explicit bounds for the linear and unitary branches

Human Compression project

This component concerns $G = \mathrm{PSL}_n(q), \mathrm{PSU}_n(q)$, $n \geq 4$. The involution lower bounds are acquired from the distinctly accepted classical lower component. The standard descriptions of their full automorphism groups, finite-field forms and classical matrix orders are retained structural inputs. We use the explicit upper bound of Lawther–Liebeck–Seitz, Proposition 1.3: $\mathrm{i}_2(\mathrm{Aut}\, S) < 2(q^{N+r} + q^{N+r-1})$ for a simple classical group of algebraic rank r and N positive roots. The exact maximal-overgroup cover stated in the source Table4 is a named retained classification premise, with its cited Bereczky/Bamberg–Penttila provenance and the acquired author erratum. Their full primary texts were not obtained; their original proofs have not been independently verified. The arithmetic below does not reprove this classification. This component recovers only the linear and unitary positive branches, not all of L045.

The chosen tori. Write $q = p^f$. In the linear case let X generate the norm-one Singer subgroup of $\mathrm{SL}_n(q)$ and let x be its projective image. In the unitary case with odd n, use the determinant-one subgroup of the irreducible unitary Singer torus. For even n, use a generator on a nondegenerate $(n-1)$-space, and its inverse determinant on the complementary line. Put $J = \mathrm{PGL}_n(q)$ or $\mathrm{PGU}_n(q)$, respectively. The resulting projective torus orders are

$$T_L = \frac{q^n - 1}{q - 1}, \qquad T_U = \begin{cases} (q^n + 1)/(q + 1) & n \text{ odd}, \\ q^{n-1} + 1 & n \text{ even}. \end{cases}$$

The element orders are these orders divided by $d = \gcd(n, q-1)$ or $\gcd(n, q+1)$, respectively. Here GU denotes the unitary isometry group.

Lemma 1. *For these elements, $C_J(x) = T$. There is no inner-diagonal inverter of x. Every involutory automorphism inverting x lies in the single pure graph coset of J. Consequently their number is at most $|T|$.*

Proof. In the linear case $M = |x|$ satisfies

$$M \geq \frac{q^n - 1}{(q-1)^2} > q^{n/2} + 1,$$

because $q^{n/2} - 1 \geq q^2 - 1 > (q-1)^2$. A projective centralizer lift normalizes the field algebra generated by X, and so acts on an eigenvalue by q^j, $0 \leq j < n$. Projective equality implies $M \mid q^j - 1$. For $j \neq 0$ this is impossible, since the gcd with $q^n - 1$ is at most $q^{n/2} - 1$. Thus the lift is field multiplication and the centralizer is exactly T. An inner inverter would instead give $M \mid q^j + 1$. Squaring gives the same contradiction unless n is even and $j = n/2$; that case is also excluded by the displayed bound.

The quotient $\mathrm{Aut}(G)/J$ is $C_f \times C_2$, with graph factor inverse transpose. An involutory reverser having nonzero field projection has f even and field exponent $f/2$. Eigenvalue comparison then gives $M \mid p^{f/2+fj} \pm 1$. Squaring implies $M \mid q^{\gcd(n, 2j+1)} - 1$. A proper gcd is at most $n/2$. Otherwise n is odd and $2j + 1 = n$, and the original divisor is at most $q^{n/2} + 1$. Both alternatives contradict the bound. The only remaining coset is the pure graph coset.

For odd unitary n, set $M = (q^n + 1)/(d(q+1))$. For $0 < j < n$, put $g = \gcd(n, j)$; then $\gcd(q^n + 1, q^{2j} - 1) = q^g + 1 < M$. Indeed, if $g = 1$, $n \geq 5$ and $q^n + 1 > (q+1)^3$. Otherwise $g \geq 3$, $n \geq 3g$, and $q^{2g} - q^g + 1 > (q+1)^2$. The same eigenvalue argument proves regularity. For even n, the unique one-dimensional eigensummand forces a scalar in projective commutation to be one, and the irreducible complementary summand then gives the stated torus centralizer. In both cases the normalizer of this torus modulo the torus has odd order (n or $n-1$): on its irreducible summand the normalizer acts by the cyclic field automorphisms. It cannot induce inversion of x. The unitary quotient $\mathrm{Aut}(G)/J$ is cyclic of order $2f$; its sole nonidentity element of order two is the q-Frobenius, or graph, coset. Finally two inverters in the same coset differ by an element centralizing x. This bounds the number by $|T|$; no assertion that all inverters are conjugate is needed. □

Lemma 2. *For every involutory inverter α, $C_G(\alpha)$ embeds in $\mathrm{PGO}_n^\epsilon(q)$ when q is odd, for some form sign, and in the isometry group $J_n(q)$ of a nonalternating symmetric bilinear form when q is even. For odd n the latter group is $\mathrm{Sp}_{n-1}(q)$.*

Proof. For PSL, the graph form B obeys $X^tB = zBX$ for a scalar z. Similarity between X and zX gives $M \mid q^j - 1$, so Lemma 1 forces $z = 1$. A bilinear form making field multiplication self-adjoint has form $B(u,v) = \mathrm{Tr}(cuv)$: write $B(u,v) = \ell(uv)$ and use the nondegenerate trace pairing. It is symmetric and, in characteristic two, nonalternating, since $u \mapsto \mathrm{Tr}(cu^2)$ is nonzero. The projective fixed group of the graph automorphism is its projective similitude group.

For PSU, take the hermitian form h linear in its first variable, and represent α by a q-semilinear anti-isometry a. Its square is cI. Commutation with this square gives $c \in \mathbb{F}_q$, and the isometry condition gives $c^{q+1} = 1$, hence $c = \pm 1$. The bilinear form $B(v,w) = h(v, aw)$ satisfies $B(w,v) = cB(v,w)$. The inverter relation makes X self-adjoint up to a scalar. As above that scalar is one: for odd n use the q^{2j} eigenvalue comparison in Lemma 1; for even n use the unique eigenline. Distinct primary summands are B-orthogonal because X is self-adjoint. Its odd-dimensional irreducible summand has a trace form, which is symmetric and nonalternating in characteristic two. Thus $c = -1$ is impossible in odd characteristic, and $a^2 = 1$ in all cases.

Elementary finite-field descent gives $V = \mathbb{F}_{q^2} \otimes W$, where W is the fixed space of a; $h|_W$ is a nondegenerate symmetric $\mathbb{F}_q$ form, nonalternating in characteristic two. A projective centralizer lift g satisfies $gag^{-1} = ta$ with $t^{q+1} = 1$. Choose u with $u^{q-1} = t$; then ug commutes with a and restricts to a similitude on W. Conversely a similitude on W can be rescaled to a hermitian isometry, since the norm map is onto. Thus the ambient fixed group is $\mathrm{PGO}(h|_W)$. In characteristic two every multiplier has a unique square root, so projective similitudes identify with $J_n(q)$. In odd dimension its canonical distinguished vector has nonzero length, and its perpendicular complement is symplectic, proving $J_n = \mathrm{Sp}_{n-1}$. □

The even-dimensional nonalternating group. For $n = 2m$ in characteristic two choose a basis z, U, w with $B(z,w) = B(w,w) = 1$, U symplectic of dimension $n-2$, and z characterized by $B(v,v) = B(v,z)^2$. Every isometry has unique parameters $A \in \mathrm{Sp}(U)$, $v \in U$, $c \in \mathbb{F}_q$:

$$z \mapsto z, \quad u \mapsto Au + B(Au, v)z, \quad w \mapsto w + v + cz.$$

These formulas preserve all pairings and describe every isometry by passing to $z^\perp/\langle z \rangle$. Squaring gives the identity exactly when $A^2 = 1$ and $Av = v$; the remaining coefficient $B(v, Av)$ is then zero. It follows, including the identity image, that

$$\mathrm{i}_2(J_n) = q^{n-1} - 1 + \sum_{A \in I_2(\mathrm{Sp}_{n-2})} q^{\dim \mathrm{Fix}(A)+1} \le q^{n-1} - 1 + q^{n-2}\mathrm{i}_2(\mathrm{Sp}_{n-2}). \tag{1}$$

For $n = 4$, $\mathrm{i}_2(\mathrm{SL}_2(q)) = q^2 - 1$ and every nonidentity involution has a one-dimensional fixed space, giving $\mathrm{i}_2(J_4) = q^4 + q^3 - q^2 - 1$.

Dimension four in odd characteristic. The determinant model on 2×2 matrices gives $\mathrm{PGO}_4^+(q) \simeq \mathrm{PGL}_2(q) \wr C_2$: the connected transformations are $X \mapsto AXB^{-1}$, and adjugation exchanges the factors up to inner automorphisms. The norm model on hermitian 2×2 matrices gives $\mathrm{PGO}_4^-(q) \simeq \mathrm{PGL}_2(q^2) \rtimes \langle \sigma \rangle$, with σ the field involution. Both maps are faithful modulo the displayed scalar kernels and have the full projective similitude orders from the ordinary orthogonal order formula. Since $\mathrm{i}_2(\mathrm{PGL}_2(q)) = q^2$, the plus involution count is $(q^2+1)^2 - 1 + |\mathrm{PGL}_2(q)|$. In the minus case every outer involution is conjugate to σ: rescale its semilinear lift to square one and apply finite-field descent. Its class has size $|\mathrm{PGL}_2(q^2)|/|\mathrm{PGL}_2(q)| = q(q^2+1)$. Thus the two counts are $q^4 + q^3 + 2q^2 - q$ and $q^4 + q^3 + q$, respectively.

An explicit uniform bound. Combining these formulas with LLS Proposition 1.3 gives $\mathrm{i}_2(C_G(\alpha)) \le \mu_n(q)$, where

n, q	$\mu_n(q)$
4, q odd	$q^4 + q^3 + 2q^2 - q$
4, q even	$q^4 + q^3 - q^2 - 1$
$n \ge 5$ odd	$2(q^{(n^2-1)/4} + q^{(n^2-1)/4-1})$
$n = 2m \ge 6$, q odd	$2(q^{m^2} + q^{m^2-1})$
$n = 2m \ge 6$, q even	$q^{2m-1} - 1 + 2q^{2m-2}(q^{m^2-m} + q^{m^2-m-1})$.

For odd q and even $n \geq 6$, the faithful embedding of projective similitudes in $\mathrm{Aut}(\mathrm{P}\,\Omega_n^\epsilon)$ uses only that a matrix centralizing the natural derived group is scalar; $D_3 = A_3$ is included. The sole nonsimple symplectic group in these bounds is $\mathrm{Sp}_4(2) \simeq S_6$, whose 75 involutions satisfy the stated numerical upper bound. No use of an automorphism theorem for a nonsimple group is made.

Explicit retained maximal-overgroup cover. Retain the published source Table4 assertion that every proper subgroup containing the chosen x lies in one of the following actual proper subgroups: in PSL_n, one extension-field group $(\mathrm{GL}_{n/r}(q^r).r) \cap G$ for each prime $r \mid n$; in odd-dimensional PSU_n, the corresponding unitary groups for each prime $r \mid n$; in even-dimensional PSU_n, the single nondegenerate $(n-1)+1$ stabilizer. The notation means the natural projective image, with its actual determinant condition. Uniqueness of the extension-field structures containing the Singer algebra explains "one" for each r. Unlike the source's definition of $\mathcal{M}(x)$, this covering family *includes* the torus normalizer when $r = n$. For composite degrees the normalizer preserves the unique prime-degree subfields and is already contained in an extension-field member. The even-unitary normalizer preserves the unique eigenline and its perpendicular complement. The element and projective orders here agree exactly with source Table3 in these two families, so the retained cover applies to these actual tori. Completeness of the cover, with the source's excluded small parameters, is the named borrowed classification; it is not inferred from these containment facts or a numerical sample. The acquired Bamberg erratum adds the odd-prime extension-field groups in odd unitary dimension, already present in the source's Table4.

Lemma 3. *Under that covering hypothesis, the following explicit number $E_n(q)$ bounds its total involutions. Put $a_k = (k^2+k-2)/2$. For even-unitary n, take*

$$E_n^U = s\{2(q^{a_{n-1}} + q^{a_{n-1}-1}) + 1\} - 1, \qquad s = \begin{cases} 1 & q \text{ even}, \\ 2 & q \text{ odd}. \end{cases}$$

In the other cases sum, over prime $r \mid n$, the quantities

$$E_{n,r} = \begin{cases} 0 & r = n, \\ (q+1)\{2(q^{2a_{n/2}} + q^{2a_{n/2}-2}) + 1\} - 1 & r = 2, \\ s\{2(q^{ra_{n/r}} + q^{r(a_{n/r}-1)}) + 1\} - 1 & r \text{ odd},\ r < n. \end{cases}$$

(The $r = 2$ case occurs only for the linear family.)

Proof. The map from the linear part of an extension-field subgroup to the smaller projective group has cyclic scalar kernel. Every fibre of a central cyclic kernel contains at most two involutions, or at most one when the kernel has odd order; include the identity image and subtract its identity lift. For odd r, all involutions lie in this linear part, and LLS applies to the smaller simple group over q^r. For $r = 2$ the image lies in $\mathrm{PGL}_{n/2}(q^2).2 \leq \mathrm{Aut}(\mathrm{PSL}_{n/2}(q^2))$; its kernel has order at most $q+1$, so use that entire fibre bound instead of assuming centrality in the field coset. For even-unitary n, projection of the $(n-1)+1$ stabilizer to $\mathrm{PGU}_{n-1}(q)$ has cyclic central kernel, giving the first formula. If $r = n$ is prime, n is odd, and the projective torus and its order-n normalizer quotient both have odd order; hence the term is zero. Every use here is in a simple smaller degree, apart from the already excluded $\mathrm{PSU}_3(2)$ case. These are upper bounds for actual subgroups, not assumed equalities between abstract group names. □

Proposition 4. *For all parameters outside the source's two finite exception lists,*

$$\mathrm{i}_2(G) > |T|\mu_n(q) + E_n(q).$$

Thus, relative to the stated retained covering classification, those linear and unitary parameters admit the generating pair required by L045.

Proof. Here are explicit estimates paying the unbounded rank range. For odd $n \geq 7$, $|T|\mu_n \leq 6q^{(n^2-1)/4+n-1}$. At $n = 7$ the covering sum is zero; the ratio to $q^{(n^2-1)/2}$ is at most $6/64 < 1/4$. For odd $n \geq 9$, each prime term is at most $6q^{n^2/6+n/2}$, and there are at most n terms. The sum of the two ratios is therefore bounded by

$$6\,2^{-(n-1)(n-3)/4} + 6n\,2^{-(2n^2-3n-3)/6} < 1/4.$$

At $n = 9$ this is $6/4096 + 54/2^{22} < 1/4$; both summands decrease as n increases by two (the second exponent increases by at least twelve). For $n = 5$, the sum is zero and, when $q \geq 4$, $|T|\mu_5 < (10/3)q^{10} < q^{12}/4$.

For $n = 2m \geq 8$, the reversing term is at most $8q^{m^2+3m-3}$. The linear covering sum is at most $6nq^{m^2+m-1}$, giving ratios at most

$$8\,2^{-(m^2-3m+3)} + 12m\,2^{-(m^2-m+1)} < 1/4;$$

at $m = 4$ their sum is $1/16 + 48/8192$, and both decrease. For the unitary covering sum in even characteristic the ratio is at most $3\,2^{-(m+1)}$; adding the reversing bound gives at most $1/16+3/32 < 1/4$ at $m = 4$, again decreasing. In odd characteristic the sharper reversing ratio is $4q^{-(m-1)^2}$ and the covering ratio is at most $6q^{-(m+1)}$; the sum is already less than $1/4$ at $m = 4, q = 3$ and decreases.

For $n = 6, q \geq 4$ the reversing term is less than $4q^{15} + (4/3)q^{10}$, while either covering sum is less than $6q^{14}$. The normalized sum is at most $4/q^3 + 6/q^4 + 4/(3q^8) < 1/4$ at $q = 4$ and decreases. For $n = 4, q \geq 7$, $\mu_4 \leq (58/49)q^4$, $|T| < (7/6)q^3$ and either covering sum is less than $5q^5$. The normalized sum is at most $29/(21q) + 5/q^3 < 1/4$ at $q = 7$. For $n = 4, q = 5$ the exact accepted class lower bounds and these upper expressions are, respectively,

G	class lower bound	$\lvert T\rvert\mu_4 + E_4$
$\mathrm{PSL}_4(5)$	251875	131825
$\mathrm{PSU}_4(5)$	170625	115171.

The unused small cases $n = 4, q = 2, 3, 4$ and $n = 5, 6, q = 2, 3$ are all included in the separately certified finite exceptions. The acquired lower theorem $\mathrm{i}_2(G) > q^{\lfloor n^2/2 \rfloor}/4$ now proves the assertion.

Finally, an involution t outside all cover members generates G with x. If an automorphism inverted x and fixed t, its square would fix both generators and hence be the identity. Thus any such obstruction is already one of the at most $|T|$ involutory inverters, whose fixed involutions have been counted. Choosing an involution outside their union gives the required pair. □

Attribution and limits. The source is Leemans–Liebeck, arXiv1603.07713v3, Section4 and Tables3–5. The new direct torus/descent proofs avoid its small-field application of Seitz Theorem2.8. The exact radical count and scalar-fibre bounds pay previously implicit interfaces. Root suggested organizing reversers as T-torsors over outer cosets; the present single-coset exclusions and linear/unitary estimates are this author's deductions. The unavailable full classification texts are not represented as acquired or reread, and the source's incorrect unitary degree-four half bound is not used. The finite35 component and classical lower theorem retain their separate source and reviewer roles.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, Section4, Tables3–5.

[2] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), Proposition1.3.

[3] A. Bereczky, *Maximal overgroups of Singer elements in classical groups*, J. Algebra 234 (2000), 187–206. Classification retained through the exact statement in [1], not an acquired full-text proof.

[4] J. Bamberg and T. Penttila, *Overgroups of cyclic Sylow subgroups of linear groups*, Comm. Algebra 36 (2008), 2503–2543. Classification retained through [1]; author correction read at `https://johnbamberg.github.io/errata.html`.

Explicit symplectic reverser and overgroup bounds

Human Compression project

This mathematical component is for $G = \mathrm{PSp}_{2m}(q)$, $m \geq 2$. The complete Singer maximal-overgroup classification is retained from Bubboloni–Spiga–Weigel Lemma5.1, citing Malle–Saxl–Weigel Theorem1.1. Its full primary statement has been acquired and read. It includes a unitary overgroup omitted from the target's Table4 and explicitly covers the low ranks; its two extra parameters are among the acquired finite exceptions. The earlier Malle–Saxl–Weigel proof itself was not obtained or reread. All source finite exceptions are acquired from the finite35 certificates. The lower bounds are acquired from the accepted classical lower theorem, and the nonalternating isometry group calculation from the companion linear/unitary proof. This proves the symplectic positive branch relative to that explicit classification premise, not the complete original disposition. Write $\nu_2(H) = \mathrm{i}_2(H) + 1$, the number of solutions of $h^2 = 1$.

1 The torus and all possible reversers

Put $E = \mathbb{F}_{q^{2m}}$, $E_0 = \mathbb{F}_{q^m}$, and $\bar{u} = u^{q^m}$. Choose $\gamma \neq 0$ with $\bar{\gamma} = -\gamma$, and take the alternating form

$$B(u, v) = \mathrm{Tr}_{E/\mathbb{F}_q}(\gamma u \bar{v}).$$

Multiplication by a generator ω of the norm-one group of E/E_0 is an isometry X of order $q^m + 1$. Its projective image x has order $M = (q^m + 1)/\gcd(2, q - 1)$. In $J = \mathrm{PGSp}_{2m}(q)$ its torus centralizer has order $D = q^m + 1$.

Here is the regularity and outer-coset check. For $m \geq 2$ in the unexcluded parameters, $M > q^{m/2} + 1$. A projective centralizer normalizes the field algebra generated by X and induces a q^j-power map on E. It follows that $M \mid q^j - 1$. For $0 < j < 2m$, the gcd with $q^m + 1$ is either at most two or is $q^{\gcd(m,j)} + 1$, where the gcd is a proper divisor of m. It is at most $q^{m/2} + 1$, a contradiction. Thus the centralizer consists of multiplication maps, and its projective similitude part is the torus of order D. The same argument applied after squaring shows that a natural projective inverter induces exactly the q^m-power map. It therefore has a lift $h : u \mapsto c\bar{u}$. This is a similitude precisely when $a = c\bar{c} \in \mathbb{F}_q^*$; its multiplier is $-a$ and its square is aI.

For $m \geq 3$, the quotient $\mathrm{Aut}(G)/J$ is the cyclic field group. A nontrivial involutory field projection has f even, $q = p^f$, and exponent $f/2$. Eigenvalue comparison would imply $M \mid p^{f/2+fj} + 1$ for some j, and hence $M \mid q^{2j+1} - 1$. But $\gcd(q^m + 1, q^{2j+1} - 1) \leq 2$, impossible. The same excludes the ordinary field involution for $m = 2$. Thus all involutory reversers are natural, except possibly the exceptional graph-field coset of $\mathrm{Sp}_4(q)$ in characteristic two. There are at most D reversers in any one coset, since their pairwise quotients centralize x.

For $\mathrm{Sp}_4(q)$ in characteristic two, the outer group is cyclic of order $2f$. If f is even its involution is a field automorphism and is excluded above. If f is odd, the additional involutory coset has fixed group $\mathrm{Sz}(q)$, by Aschbacher–Seitz Section19, (19.3)–(19.5). These exact outer-automorphism statements, with the 1978 erratum retained, are borrowed structural inputs. We may uniformly count at most $2D$ reversers in degree four and characteristic two; all other cases use D.

2 Their fixed groups and involutions

Suppose first that q is odd. Since $h^2 = aI$ and its multiplier is $-a$, h is skew-adjoint for B. If a is a square, its two m-dimensional eigenspaces are paired totally isotropic spaces, and the isometric centralizer is $\mathrm{GL}_m(q)$. If a is nonsquare, the algebra $\mathbb{F}_q[h]$ is $\mathbb{F}_{q^2}$ and the same pairing gives a hermitian form; the centralizer is $\mathrm{GU}_m(q)$. Projective commutation permits a lift to send h to $-h$, so $C_G(\alpha)$ is contained in the corresponding normalizer, modulo $\langle -I \rangle$, with quotient of order at most two over the linear or unitary part. For $m \geq 3$, its image on the derived projective group is a subgroup of $\mathrm{Aut}(\mathrm{PSL}_m(q))$ or $\mathrm{Aut}(\mathrm{PSU}_m(q))$. The kernel is cyclic of order at most $q + 1$. The extra coset acts as inverse transpose, a genuine outer graph map in these degrees; it introduces no further kernel. Lawther–Liebeck–Seitz Proposition1.3 consequently gives

$$\mathrm{i}_2(C_G(\alpha)) \leq (q + 1)\{2(q^{a_m} + q^{a_m - 1}) + 1\} - 1, \qquad a_m = (m^2 + m - 2)/2. \tag{1}$$

The entire kernel size is charged: it is not assumed central in the normalizing coset.

For q even, rescale a natural inverter to have square one. Hilbert90 for E/E_0 gives the required conjugating multiplier. Multiplying that multiplier by an element of E_0 makes its norm one, since squaring is onto in E_0; thus the conjugation is by a torus isometry and takes the inverter to $u \mapsto \bar{u}$. Its square-zero part has rank m. The induced form is nonalternating, since $B(u, \bar{u}) = \mathrm{Tr}_{E/\mathbb{F}_q}(\gamma u^2)$ is not identically zero. In a symplectic basis it is therefore

$$t = \begin{pmatrix} I & I \\ 0 & I \end{pmatrix}.$$

Every commuting isometry has diagonal blocks A, A, where $AA^t = I$, and its off-diagonal block can be written DA^{-t} for a symmetric matrix D. Equivalently the centralizer is

$$\mathrm{Sym}_m(q) \rtimes J_m(q), \qquad A : D \longmapsto ADA^t,$$

where J_m is the isometry group of a nonalternating symmetric form. This follows directly from the two block commutation equations and the symplectic pairing equation; the kernel is the additive space of symmetric matrices, of order $q^{m(m+1)/2}$. In particular

$$\mathrm{i}_2(C_G(\alpha)) \le q^{m(m+1)/2}\nu_2(J_m(q)) - 1. \tag{2}$$

The identity image is included. The acquired elementary description gives $J_m = \mathrm{Sp}_{m-1}$ for odd m and

$$\nu_2(J_m) \le q^{m-1} + q^{m-2}\mathrm{i}_2(\mathrm{Sp}_{m-2}) \quad (m \text{ even}).$$

For $m = 2$, J_2 is the additive group of order q. Each nonidentity $A \in J_2$ fixes a two-dimensional subspace of the three-dimensional space $\mathrm{Sym}_2(q)$: conjugate its single Jordan block to $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and solve $ADA^t = D$. Thus the exact centralizer involution count is

$$\mu_2(q) = q^3 + (q-1)q^2 - 1 = 2q^3 - q^2 - 1. \tag{3}$$

In the exceptional outer coset, the Suzuki group has one involution class, with centralizer q^2, so its involution count is $(q-1)(q^2+1) < \mu_2(q)$. Its class order is retained from the primary involution inventory. This proves the same bound for every reverser.

The odd-characteristic degree-four case. Here the graph on the derived degree-two group is inner, so use an exact count rather than the kernel assertion in (1). In the split case take

$$\mathrm{GL}_2(q) : A \mapsto \mathrm{diag}(A, A^{-t}), \qquad w = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

Then $w^2 = -I$ and w acts on A by A^{-t}. In the nonsplit case the same normalizer is described by $\mathrm{GU}_2(q)$ and a semilinear w with square $-I$, acting by conjugation on the field entries. Such a w is obtained by multiplying coordinate conjugation by a scalar of norm -1. Modulo $\langle -I \rangle$, an element of the linear part is an involution when $A^2 = I$ or $-I$. Splitting their two eigenspaces, or using the irreducible quadratic centralizer when needed, gives

	$q \equiv 1 \pmod 4$	$q \equiv 3 \pmod 4$
$\mathrm{i}_2(\mathrm{GL}_2/\{\pm I\})$	$q(q+1)+1$	q^2
$\mathrm{i}_2(\mathrm{GU}_2/\{\pm I\})$	q^2	$q(q-1)+1.$

In the other coset, square $\pm I$ means $A = \pm A^t$. There are $q^2(q-1)$ nonsingular symmetric and $q-1$ nonsingular alternating linear matrices, hence $(q-1)(q^2+1)/2$ projective solutions. For unitary matrices, symmetric ones correspond by finite-field descent to the two orthogonal forms, so their number is $|\mathrm{GU}_2|(1/|O_2^+| + 1/|O_2^-|) = q^2(q+1)$; there are $q+1$ alternating ones. This gives $(q+1)(q^2+1)/2$ solutions. Consequently, for $q \ge 5$,

$$\mathrm{i}_2(C_G(\alpha)) \le \tfrac{1}{2}(q+1)(q^2+1) + q^2 + q + 1 < q^3. \tag{4}$$

The inequality reduces to $q^3 - 3q^2 - 3q - 3 > 0$, already positive at five and increasing thereafter.

3 Retained covering and the uniform inequalities

Retain the complete list of maximal overgroup *types* from Bubboloni–Spiga–Weigel Lemma5.1 for the full Singer lift X in $\mathrm{Sp}_{2m}(q)$, $m \geq 2$, $(m,q) \neq (2,2)$: the extension-field types $\mathrm{Sp}_{2m/r}(q^r).r$ for primes $r \mid m$; additionally $O^-_{2m}(q)$ when q is even; and additionally $\mathrm{GU}_m(q).2$ when m, q are odd. Here $\mathrm{GU}_m(q)$ is the isometry group on the m-space over $\mathbb{F}_{q^2}$. The only extra cases in that statement are $(m,q) = (4,2)$ with a $\mathrm{PSL}_2(17)$ overgroup and $(m,q) = (2,3)$ with a $2^{1+4}_-.A_5$ overgroup. Both are paid by the acquired finite35 certificates, so no uniform inequality is applied there. The unitary type is absent from the target Table4, so that printed list is not used as a complete cover. Passing to projective groups causes no new case: the full inverse image of a subgroup containing x contains X and the scalar centre.

Here are the multiplicities for the fixed element, rather than just the conjugacy types. In an extension-field normalizer, X^r is linear over $K = \mathbb{F}_{q^r}$. Its order is $(q^m+1)/\gcd(r, q^m+1)$ and is greater than $q^{m/2}+1$ in all unexcluded parameters. For $m \geq 6$ use $q^{m/2} - 1 > m$; for $m = 2, 3, 4$ direct substitution at the smallest retained q proves the inequality. For $m = 5$, the only remaining smallest case is $q = 2$, where $r = 5$ does not divide 33. The same eigenvalue gcd argument as above proves that X^r is irreducible. Thus $K \subseteq C_{\mathrm{End}_{\mathbb{F}_q}(E)}(X^r) = E$, where a subfield of this order is unique. The full normalizer of this one field structure is one subgroup, so there is at most one extension-field term for each r. For the unitary type, apply the same argument to X^2. Its unique quadratic field is the subfield $\mathbb{F}_{q^2} \subset E$; since m is odd, the bar map restricts to its nontrivial involution. The alternating trace form determines the hermitian form, and the full normalizer of this field structure is the one unitary normalizer. In characteristic two, two X-invariant quadratic forms polarizing to B differ by the square of a linear functional fixed by X. Irreducibility excludes a nonzero such functional. Hence the orthogonal overgroup is unique as well. The normalizer of $\langle X \rangle$ preserves each of the unique prime-degree extension fields, so it is included even when m is prime. No uncharged normalizer term is discarded.

For the extension-field term, the actual scalar quotient embeds in the automorphism group of $\mathrm{PSp}_{2m/r}(q^r)$. For odd r, its involutions lie in the linear part, but the same upper bound suffices. The bound is

$$E_r < 2\{q^{m^2/r+m} + q^{m^2/r+m-r}\}.$$

The additional even-characteristic orthogonal term is bounded by $2(q^{m^2} + q^{m^2-1})$; its isometry group embeds in $\mathrm{Aut}(\Omega^-_{2m}(q))$. This includes $m = 3$ through $D_3 = A_3$ and $m = 2$ through $O^-_4(q) = \mathrm{SL}_2(q^2).2$. For the additional odd-characteristic unitary normalizer, the projective image acts on $\mathrm{PSU}_m(q)$ with cyclic kernel of order at most $q+1$; its extra coset is a graph automorphism for $m \geq 3$. Thus its entire involution contribution is at most the right side of (1). All smaller groups used here are simple. Thus, with at most m prime terms, their total is at most $3mq^{m^2/2+m}$; for $m = 3$ it is at most $3q^6$. In characteristic two add $3q^{m^2}$ for the orthogonal term.

For odd q and $m \geq 3$, the reversing contribution from (1) together with that unitary term when present is less than $5q^{(m^2+3m)/2}$. Indeed, charge $(q^m+2)\mu$ uniformly, and use $q^m + 2 \leq (29/27)q^m$, $q+1 \leq 4q/3$, and $2(q^{a_m} + q^{a_m-1}) + 1 \leq 3q^{a_m}$. Relative to the acquired lower scale q^{m^2+m}, at $m = 3$ the total is at most $5/q^3 + 3/q^6 < 1/4$ for $q \geq 3$. For $m \geq 4$ it is at most

$$5\,3^{-m(m-1)/2} + 3m\,3^{-m^2/2} < 1/4.$$

Both summands decrease from $m = 4$, where their sum is less than 0.009. The accepted quarter lower bound proves the strict counting inequality.

For q even the accepted explicit class lower bound is stronger: $\mathrm{i}_2(G) > (2/3)q^{m^2+m}$. For odd $m \geq 5$, (2) and LLS give reversing contribution less than $5q^{(3m^2+6m-1)/4}$. Its ratio, with the other terms, is at most

$$5\,2^{-(m-1)^2/4} + 3\,2^{-m} + 3m\,2^{-m^2/2} < 2/3.$$

At $m = 5$ the sum is less than 0.41 and all terms decrease. For even $m \geq 6$, the bound $\nu_2(J_m) < 4q^{m^2/4+m/2-2}$ follows from its displayed formula and LLS. The ratio is instead at most

$$5\,2^{-(m^2/4-m+2)} + 3\,2^{-m} + 3m\,2^{-m^2/2} < 2/3;$$

at $m = 6$ it is less than 0.204, again decreasing. The numerical LLS bound for the nonsimple $\mathrm{Sp}_4(2)$ is paid by its exact 75 involutions; no simplicity premise is used there.

For $m = 3$ in characteristic two, $\nu_2(J_3) = q^2$, so the reversing term is $(q^3+1)(q^8-1)$. At $q \geq 4$ the total ratio is bounded by $(1+q^{-3})/q + 2(1+q^{-1})/q^3 + 3/q^6 < 2/3$. For $m = 4$, the exact $\nu_2(J_4) = q^4 + q^3 - q^2$ bounds the

reversing term by $5q^{18}$. At $q \geq 4$ the total ratio is at most $5/q^2 + 3/q^4 + 12/q^8 < 2/3$. The unused $q = 2$ cases are finite35 inputs.

For $m = 2$, odd $q \geq 5$, the accepted class lower bound exceeds $q^5(q-1)/2$. Equations (4) and the single extension-field bound give total less than $q^5 + 2q^4 + q^3 + 2q^2$, which is smaller already at $q = 5$; after division by q^5, the left side $(q-1)/2$ increases and the right side decreases. For even $q \geq 8$, use the exact class lower bound $(q^2-1)(q^4-1)$ and the at most $2D$ reversers. Here each of the two overgroups is an $A_1(q^2)$ automorphism subgroup: $\mathrm{Sp}_2(q^2).2$ and $O_4^-(q) = \mathrm{SL}_2(q^2).2$. Thus LLS bounds each by $2(q^4 + q^2)$, sharpening the general orthogonal bound used above. The difference is at least

$$q^6 - 4q^5 - 3q^4 - 4q^3 - q^2 + 3 > 0.$$

Divide by q^6 and drop its positive last term; the resulting lower bound is positive at eight and increasing. The other degree-four parameters are in finite35, with the nonsimple $\mathrm{Sp}_4(2)$ excluded from the original positive simple-group claim.

An involution outside the covered proper subgroups generates G with x. Any reversing automorphism fixing this partner squares to the identity, so all such obstructions have been included. The strict inequalities therefore prove the symplectic existence assertion relative to the explicit retained covering classification, without the source's unexpanded arithmetic or its uniform single-inverter-class assertion.

Attribution and limits. The target is Leemans–Liebeck, arXiv1603.07713v3, Section4. The missing unitary overgroup was identified by root from BLW Lemma3.1 before this component was frozen; this author acquired the full all-rank BSW Lemma5.1 to pay the remaining low-rank interface. Earlier drafts and their arithmetic are retained. The projective coset organization was suggested by root; the present symplectic trace-form, field exclusion, exact degree-four fibres and inequalities are this author's new derivations. AS1976 Section19 and its 1978 erratum supply the exceptional outer fixed groups; BT Proposition2.3/Table5 supplies the Suzuki class count; LLS2002 Proposition1.3 is the numerical upper theorem. The finite35 and classical lower components retain their separate authorship and distinct reviews. This is neither a proof of maximal subgroup classification nor a whole-source admission.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, Section4, Tables3–5.

[2] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), Proposition1.3.

[3] D. Bubboloni, M. S. Lucido and Th. Weigel, *2-Coverings of classical groups*, arXiv:1102.0660, Lemma3.1; there citing G. Malle, J. Saxl and T. Weigel, *Generation of classical groups*, Geom. Dedicata 49 (1994), 85–116, Theorem1.1.

[4] D. Bubboloni, P. Spiga and Th. Weigel, *Normal* 2*-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Lemma5.1, physical page40. The stated classification is retained; its earlier MSW proof was not obtained.

[5] M. Aschbacher and G. M. Seitz, *Involutions in Chevalley groups over fields of even order*, Nagoya Math. J. 63 (1976), 1–91, Section19; with the authors' 1978 corrections.

[6] T. C. Burness and A. R. Thomas, *On the involution fixity of exceptional groups of Lie type*, arXiv:1707.03562v2, Proposition2.3/Table5.

Involutions outside the reversal and subgroup obstructions

Human Compression project

Theorem 1. *Let G be a finite simple exceptional group of Lie type, other than $G_2(2)' \cong \mathrm{PSU}_3(3)$ and ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$. There are $x, t \in G$ such that*

$$|t| = 2, \qquad \langle x, t\rangle = G, \qquad \nexists \alpha \in \mathrm{Aut}(G) : \ (x^\alpha, t^\alpha) = (x^{-1}, t).$$

This is the exceptional-group part of the positive existence theorem of Leemans and Liebeck [1]. The already reviewed finite certificates cover $G_2(3)$, $G_2(4)$ and the Tits group ${}^2F_4(2)'$. The argument below treats every remaining parameter. The passage from these generators to an abstract chiral polyhedron is the retained Schulte–Weiss reconstruction theorem. In a nonabelian simple group the cyclic-subgroup intersection required there is automatic: $\langle x\rangle \cap \langle x^{-1}t\rangle$ centralizes both generators, and is therefore trivial.

1 A counting criterion with the outer automorphisms included

For a finite group H, write $i_2(H)$ for its number of elements of order two and $\nu_2(H) = i_2(H) + 1$. Thus ν_2 counts solutions of $h^2 = 1$, including the identity. If $K \lhd H$, then

$$\nu_2(H) \le |K|\nu_2(H/K). \tag{1}$$

Indeed, each solution in H maps to a solution in the quotient, and a fibre contains at most $|K|$ elements. If K is central of odd order, each such fibre contains at most one solution: any two differ by a central element whose square is one. We will use both forms, keeping the identity in every estimate involving an extension.

Lemma 2. *Let G be centreless, let $G \le J \lhd A = \mathrm{Aut}(G)$, and suppose $|x| > 2$ and $C_J(x) = T$. Put*

$$\kappa = |\{aJ \in A/J : (aJ)^2 = J\}|.$$

Suppose μ bounds $i_2(C_G(\alpha))$ for every involution $\alpha \in A$ which sends x to x^{-1}. Suppose every proper subgroup containing x lies in either $N_G(\langle x\rangle)$ or one of the listed maximal subgroups $M_1, \dots, M_s$. If

$$i_2(G) > \kappa|T|\mu + |N_G(\langle x\rangle)| + \sum_{j=1}^{s} i_2(M_j), \tag{2}$$

then x has an involutory partner as in Theorem 1.

Proof. Fix a coset of J containing a reverser a. Every other reverser in that coset differs from a by an element of $C_J(x)$; this follows by applying their quotient to x. The set of reversers in that coset thus has at most $|T|$ elements. An involutory reverser has image of square one in A/J, so there are at most $\kappa|T|$ of them in total. The first term of (2) consequently bounds the involutions fixed by any involutory reverser. The remaining terms bound all involutions which fail to generate G with x. Choose t outside this union. If an automorphism reverses x and fixes t, its square fixes the generating pair and is the identity. It is not the identity, because $|x| > 2$. It would therefore be one of the excluded involutory reversers, a contradiction. □

The source uses the corresponding single-coset estimate. Our form only requires the centralizer inside J; field and graph automorphisms are counted explicitly. It also charges the torus normalizer explicitly.

2 The tori and the retained subgroup classification

Let $\mathbf{G}$ be the adjoint algebraic group, let F be its defining endomorphism, and put $J = \mathbf{G}^F$, $G = J'$. Choose the cyclic torus $T_0 = \langle x\rangle$ in Table 1. Its existence and cyclicity are the torus results collected in

G	D	other maximal overgroups of T_0
$E_8(q)$	$\Phi_{30}(q)$	none
$E_7(q),\ q \geq 3$	$\Phi_{18}(q)(q+1)$	one ${}^2E_6(q).(q+1)/d$
$E_7(2)$	129	one $\mathrm{PSU}_8(2).2$
$E_6(q)$	$\Phi_9(q)$	one $\mathrm{PSL}_3(q^3).3$
${}^2E_6(q)$	$\Phi_{18}(q)$	one $\mathrm{PSU}_3(q^3).3$
$F_4(q),\ q \geq 3$	$\Phi_{12}(q)$	at most two ${}^3D_4(q).3$
$F_4(2)$	17	two $\mathrm{Sp}_8(2)$
$G_2(q),\ q \geq 5$	$\Phi_6(q)$	one $\mathrm{SU}_3(q).2$
${}^3D_4(q)$	$\Phi_{12}(q)$	none
${}^2F_4(q),\ q \geq 8$	$q^2 + \sqrt{2q^3} + q + \sqrt{2q} + 1$	none
${}^2G_2(q),\ q \geq 27$	$q + \sqrt{3q} + 1$	none
${}^2B_2(q),\ q \geq 8$	$q + \sqrt{2q} + 1$	none

Table 1: The chosen tori and their other maximal overgroups.

Kantor–Seress [3, Section 2]. The index of T_0 in the corresponding torus T of J is $d = (2, q-1)$ for E_7, $d = (3, q-\epsilon)$ for E_6^ϵ, and $d = 1$ otherwise. In the table D is the exact order of T.

The overgroup information, including the numbers of overgroups containing the chosen torus rather than just their conjugacy types, is the retained classification statement summarized in [1, Table 2]. Its cited source is Guralnick–Kantor [2, Propositions 6.1–6.2 and their proofs], which in turn uses the subgroup classification. Their Table III gives the ordinary-family counts; the displayed types are recorded explicitly in the summary in [1]. The small-case Table IV of [2] supplies $F_4(2)$ and $E_7(2)$. Central quotients in Table 1 are understood in their natural embeddings; the estimates below allow a covering kernel wherever needed. We are not reproving the classification of maximal subgroups.

We do prove the centralizer interface required for Lemma 2. This avoids applying a general torus-centralizer statement outside its stated field-size range.

Lemma 3. *For the choices in Table 1, $C_{\mathbf{G}}(x) = \mathbf{T}$ is the connected algebraic torus containing T, and $C_J(x) = T$. Moreover,*

$$|N_G(\langle x \rangle)| \leq |W| D, \tag{3}$$

where W is the absolute Weyl group.

Proof. Let Q be the character lattice of the adjoint torus, which is the root lattice. For an ordinary twisted or untwisted group the endomorphism on this lattice is $F^* = qu$, with u an isometry. A root α trivial on T_0 gives a character of T/T_0, and hence

$$d\alpha = (qu - I)v \quad \text{for some } v \in Q.$$

Consequently

$$\|v\| \leq \frac{d\|\alpha\|}{q-1}. \tag{4}$$

Equality requires $uv = v$. In the simply laced cases all roots have the length of a shortest nonzero lattice vector. The chosen torus types have no eigenvalue one: their characteristic polynomials are Φ_{30}, $\Phi_{18}\Phi_2$ (or $\Phi_{14}\Phi_2$ for $E_7(2)$), Φ_9, Φ_{18} and Φ_{12}, respectively. For $E_8, E_7, E_6^\epsilon, {}^3D_4$ the right side of (4) is at most this minimum length, except for ${}^2E_6(2)$. If equality could occur, the absence of a fixed vector makes the actual inequality strict. Thus no root is trivial on T_0. For $F_4(q)$ with $q \geq 3$ and $G_2(q)$ with $q \geq 5$, the ratios of longest to shortest root length are $\sqrt{2}$ and $\sqrt{3}$, so (4) is again strictly below the minimum lattice length.

For the Suzuki and Ree endomorphisms, $F^*/\sqrt{q}$ is an isometry. Replacing $q-1$ by $\sqrt{q}-1$ proves the same assertion: at the smallest allowed parameters the relevant upper ratios are $\sqrt{2}/(\sqrt{8}-1) < 1$ and $\sqrt{3}/(\sqrt{27}-1) < 1$. This includes ${}^2F_4(8)$.

The two remaining lattice checks are finite. We use simple-root bases with the E_6 diagram edges $(1,2), (2,3), (3,4), (4,5), (3,6)$ and the F_4 Cartan matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

The supplied certificate constructs all 72 and 48 roots by their simple reflections. It supplies a Weyl matrix w of characteristic polynomial Φ_9 in E_6 and one of polynomial Φ_8 in F_4. For every root, respectively,

$$(-2w - I)^{-1}(3\alpha) \notin \mathbb{Z}^6, \qquad (2w - I)^{-1}\alpha \notin \mathbb{Z}^4.$$

The determinants have absolute values 57 and 17. These are the required two torus types, and the displayed tests prove regularity in ${}^2E_6(2)$ and $F_4(2)$. Thus in all cases $C_{\mathbf{G}}(x)^\circ = \mathbf{T}$.

We check connectedness as well. For a central isogeny $\widetilde{\mathbf{G}} \to \mathbf{G}$ from the simply connected group, the component group of a semisimple centralizer embeds in the central kernel: an element centralizing x takes a lift $\tilde{x}$ to $z\tilde{x}$, and the kernel of this map is connected. If $|x|$ is coprime to the order of the kernel, then $z^{|x|} = 1$ forces $z = 1$. This applies to E_6^ϵ because $\Phi_9(q)/d$ and $\Phi_{18}(q)/d$ are prime to three; when three divides the numerator its valuation is exactly one. It applies to ${}^3D_4$ because its torus has odd order and the covering kernel is a two-group. The other cases have trivial central kernel on geometric points, except for odd-characteristic E_7.

For E_7, the exact Weyl calculation gives $C_W(u) = \langle u \rangle$ of order 18, or 14 at $q = 2$. In the rational normalizer u acts on T by the inverse of the q-power map. For $q \geq 3$, $\Phi_{18}(q)$ divides $|T_0|$, and q has order 18 modulo $\Phi_{18}(q)$: its order divides 18, $q^9 \equiv -1$, and $q^6 \equiv q^3 - 1 \not\equiv 1$ modulo that number. These exclude every proper divisor of 18. At $q = 2$, the order of two modulo 129 is 14. Thus no nonidentity Weyl-normalizer element fixes x. If its algebraic centralizer were disconnected, its component group would have order two. Frobenius acts trivially on that group, and Lang's theorem for the connected torus would give a rational representative of its nontrivial component. That contradicts the faithful normalizer action. The centralizer is therefore connected here too. The certificate checks both Weyl centralizers, not merely the orders of chosen elements.

Finally, a normalizer of $\langle x \rangle$ normalizes its algebraic centralizer $\mathbf{T}$. Its quotient by its intersection with $\mathbf{T}$ embeds in W, while that intersection has at most D elements. This proves (3). □

3 Involutions which reverse the torus generator

Write $q = p^f$. The standard structure of outer automorphisms gives the following bounds on the square-one elements of $\mathrm{Aut}(G)/J$:

G	E_8	E_7	E_6^ϵ	F_4	G_2	${}^3D_4$	${}^2F_4$	${}^2G_2$	${}^2B_2$
κ	2	2	4	2	2	2	1	1	1.

The quotient is cyclic apart from the untwisted E_6 graph factor, where it is $C_f \times C_2$. In the exceptional graph-field cases F_4 in characteristic two and G_2 in characteristic three it is cyclic of order $2f$. For ${}^3D_4$ it is cyclic of order $3f$, so $\kappa = 1$ at $q = 2, 3$. In the three Suzuki–Ree families the field degree is odd and there is no nontrivial element of square one.

Suppose first that an inner-diagonal involution reverses x. For every type other than E_6, the longest Weyl element is -1. An algebraic representative n of it can be chosen to have order two; all elements in $\mathbf{T}n$ then have order two and are conjugate over the algebraic closure, since squaring on the torus is surjective. There is a rational representative by Lang's theorem for $\mathbf{T}$. If a is any other rational reverser, an^{-1} centralizes x and hence lies in T, by Lemma 3. Thus all such reversers belong to this algebraic involution class.

For E_6 the Weyl centralizer of the Φ_9 torus element has order nine, as checked in the root certificate. The twisted torus uses its negative; its Weyl centralizer is the same group. Since the torus order is odd, $N_J(T)$ has odd order. There is consequently no inner-diagonal involutory reverser in this case.

We identify this class before using its centralizer. Let $\mathbf{B}$ be a Borel subgroup containing $\mathbf{T}$, and let N be the number of positive roots. Since n inverts $\mathbf{T}$, it interchanges $\mathbf{B}$ and its opposite $\mathbf{B}^-$. Consequently

$$C_{\mathbf{G}}(n) \cap \mathbf{B} \subseteq \mathbf{B} \cap \mathbf{B}^- = \mathbf{T}, \qquad C_{\mathbf{G}}(n) \cap \mathbf{B} = \mathbf{T}[2].$$

The last group scheme is finite, including in characteristic two. The orbit of $\mathbf{B}$ under $C_{\mathbf{G}}(n)$ in the flag variety $\mathbf{G}/\mathbf{B}$ therefore has dimension $\dim C_{\mathbf{G}}(n)$. It follows that

$$\dim C_{\mathbf{G}}(n) \leq N. \tag{5}$$

This argument uses the dimension of an orbit and its stabilizer; it requires neither a smooth stabilizer nor a separable orbit map.

The involution classification remains a retained input. The complete lists in [5, Proposition 2.3, Table 5 and Remark 2.4], with the centralizer structures of [6, Sections 13–18] and the corrections [7], give the following dimension inventory. Repeated entries may denote different rational classes.

type	N	odd characteristic	even characteristic
E_8	120	$120, 136$	$190, 156, 136, 120$
E_7	63	$69, 63, 79$	$99, 81, 79, 69, 63$
F_4	24	$24, 36$	$36, 36, 30, 24$
G_2	6	6	$8, 6$
${}^3D_4$	12	12	$18, 12$
${}^2F_4$	24	$--$	$30, 24$
${}^2G_2$	6	6	$--$
${}^2B_2$	4	$--$	4

For ordinary Frobenius endomorphisms these dimensions are the degrees of the classified centralizer order polynomials. For Suzuki–Ree endomorphisms the normalized scaling is $\sqrt{q}$, so the dimensions are twice those degrees: the two ${}^2F_4$ degrees are $15, 12$, and the ${}^2G_2$ and ${}^2B_2$ degrees are 3 and 2. These degrees come from the uniform algebraic centralizer structures, not an isolated finite group order. In the triality case the listed classes are precisely those descending to ${}^3D_4(q)$. In odd-characteristic E_7, both rational forms of the algebraic $A_7.2$ and $\mathbf{T}_1E_6.2$ classes must be allowed in J; Remark 2.4 of [5] records this distinction. In each row only the class used in Table 2 satisfies (5). The two rational A_7 forms have the same subsequent upper bound. This proves the required identification without assuming that the reverser lies in the simple subgroup. For a covering group or an adjoint group the indicated small central kernels are allowed. The powers in brackets denote the order of the normal unipotent subgroup.

type	odd characteristic	even characteristic
E_8	D_8	$[q^{84}].\,\mathrm{Sp}_8(q)$
E_7	$A_7^{\epsilon}.2$	$[q^{42}].\,\mathrm{Sp}_6(q)$
F_4	$(A_1C_3).2$	$[q^{18}].\,\mathrm{SL}_2(q)^2$
G_2	$(A_1A_1).2$	$[q^3].\,\mathrm{SL}_2(q)$
${}^3D_4$	$(A_1(q)A_1(q^3)).2$	$[q^9].\,\mathrm{SL}_2(q)$
${}^2F_4$	—	$[q^9].\,\mathrm{SL}_2(q)$
${}^2G_2$	$\mathrm{PSL}_2(q) \times 2$	—
${}^2B_2$	—	a Sylow two-subgroup

Table 2: Retained centralizer types for a lift of the inverting Weyl element.

To estimate their involutions we retain the uniform theorem of Lawther–Liebeck–Seitz [4, Proposition 1.3]:

$$i_2(\mathrm{Aut}(H)) < 2(q^{M_H} + q^{M_H-1}). \tag{6}$$

Here $M_H = N + r$ for the absolute root system, and half that value for a Suzuki–Ree group. Since the upper bound is an integer, the same expression is an upper bound for $\nu_2(\mathrm{Aut}(H))$. This observation is needed in using (1).

For E_8 in odd characteristic a central kernel of order at most four reduces the centralizer to a subgroup of $\mathrm{Aut}(P\Omega^+_{16}(q))$. Thus $\nu_2 \le 8(q^{64} + q^{63})$, which is below $2q^{84}(q^{20} + q^{19})$. For odd E_7, a central kernel of order at most eight gives $\nu_2 \le 16(q^{35} + q^{34})$, below $2q^{42}(q^{12} + q^{11})$. In even characteristic (1) gives the latter two expressions directly.

For odd q, $\nu_2(\mathrm{PGL}_2(q)) = q^2 + 1$. The central products in the next three rows have kernel at most two, and their quotients embed in the products of the corresponding inner-diagonal factors. They give

$$\begin{array}{c|c} F_4 & 4(q^2+1)(q^{12}+q^{11}) < q^{22} \\ G_2 & 2(q^2+1)^2 < q^5 \\ {}^3D_4 & 2(q^2+1)(q^6+1) < q^{11}. \end{array}$$

These inequalities hold for odd $q \ge 3$. In even characteristic $\nu_2(\mathrm{SL}_2(q)) = q^2$; multiplication by the indicated unipotent orders gives q^{22}, q^5, q^{11}, and also q^{11} for ${}^2F_4$. For the small Ree group, $i_2(\mathrm{PSL}_2(q)\times 2) = q(q-1)+1 < 2q^2$. For the Suzuki group the acquired Sylow-subgroup calculation gives $i_2(C_G(n)) = q - 1$.

The outer involutions must still be included. Proposition 1.1 of [4] gives their fixed groups. A field involution has a group of the same type over s with $s^2 = q$, allowing inner-diagonal extensions. Formula (6) bounds its involutions by $4q^{M_G/2}$. This fits the preceding bounds for $E_8, E_7, F_4, G_2, {}^3D_4$. For E_6^{ϵ} a field or

graph-field involution is bounded by $4q^{21}$. A graph involution has fixed group $F_4(q)$, $C_4(q)$, or an involution centralizer in $F_4(q)$; in every case $2(q^{28}+q^{27})$ is sufficient. The exceptional graph-field involutions of F_4 and G_2 have fixed groups ${}^2F_4(q)$ and ${}^2G_2(q)$ respectively. Their bounds $2(q^{14}+q^{13})$ and $2(q^4+q^3)$ fit q^{22} and q^5. At $F_4(2)$ the fixed group is ${}^2F_4(2)$, which embeds by conjugation in the automorphism group of its simple derived Tits group; Proposition 1.3 of [4] still gives the bound $49152 < 2^{22}$. No field involution occurs in the three Suzuki–Ree families themselves.

We have therefore proved that the quantities C in Table 4 bound $i_2(C_G(\alpha))$ for *every* involutory reverser, not only the inner-diagonal ones.

4 Lower bounds from one explicit conjugacy class

We also make the constants in the lower bounds explicit. Let J again denote the inner-diagonal group. Proposition 2.3, Remark 2.2 and Table 5 of [5] provide the involutions in Table 3. Each lies in G, and its J-class is its G-class. Thus $|J|/|C_J(t)|$ is a lower bound for $i_2(G)$. These are retained class orders, not estimates inferred from asymptotic notation.

type	$\lvert C_J(t)\rvert$, odd characteristic	$\lvert C_J(t)\rvert$, even characteristic
E_8	$\lvert \mathrm{SO}^+_{16}(q)\rvert$	$q^{84}\lvert \mathrm{Sp}_8(q)\rvert$
E_7	$2\lvert \mathrm{SL}^\epsilon_8(q)\rvert$, $q\equiv\epsilon \pmod 4$	$q^{42}\lvert \mathrm{Sp}_6(q)\rvert$
E_6^ϵ	$\lvert \mathrm{SL}_2(q)\rvert\,\lvert \mathrm{SL}^\epsilon_6(q)\rvert$	$q^{27}\lvert \mathrm{SL}_2(q)\rvert\,\lvert \mathrm{SL}^\epsilon_3(q)\rvert$
F_4	$\lvert \mathrm{SL}_2(q)\rvert\,\lvert \mathrm{Sp}_6(q)\rvert$	$q^{18}\lvert \mathrm{SL}_2(q)\rvert^2$
G_2	$\lvert \mathrm{SL}_2(q)\rvert^2$	$q^3\lvert \mathrm{SL}_2(q)\rvert$
${}^3D_4$	$\lvert \mathrm{SL}_2(q)\rvert\,\lvert \mathrm{SL}_2(q^3)\rvert$	$q^9\lvert \mathrm{SL}_2(q)\rvert$
${}^2F_4$	—	$q^9\lvert \mathrm{SL}_2(q)\rvert$
${}^2G_2$	$\lvert \mathrm{SL}_2(q)\rvert$	—
${}^2B_2$	—	q^2

Table 3: Centralizer orders of the selected lower-bound classes.

Substitution of the standard group-order products gives

$$i_2(G) > q^a/c, \qquad \begin{array}{c|ccccccccc} G & E_8 & E_7 & E_6^\epsilon & F_4 & G_2 & {}^3D_4 & {}^2F_4 & {}^2G_2 & {}^2B_2 \\ \hline a & 128 & 70 & 40 & 28 & 8 & 16 & 14 & 4 & 3 \\ c & 2 & 4 & 2 & 2 & 2 & 2 & 2 & 2 & 2. \end{array} \tag{7}$$

Here is an exact way to check every constant. Form the integer polynomial $P(q) = c|J| - q^a|C_J(t)|$ from the row in Table 3. Expand $P(z+q_0)$, where q_0 is the smallest allowed parameter of that parity. In all eighteen cases every coefficient is nonnegative and the constant coefficient is positive. The certificate stores every coefficient, using integer arithmetic; this proves positivity for all real $q \ge q_0$ and hence for every required prime power. The order products used are

$$\lvert \mathrm{SL}^\epsilon_n(q)\rvert = q^{n(n-1)/2}\prod_{i=2}^{n}(q^i-\epsilon^i), \quad \lvert \mathrm{Sp}_{2m}(q)\rvert = q^{m^2}\prod_{i=1}^{m}(q^{2i}-1),$$

and the analogous displayed exceptional order products in the certificate. The correction to the ${}^2E_6$ centralizer in [7] is respected. For the Suzuki groups we will use the stronger exact value $(q-1)(q^2+1)$, obtained from their single involution class.

5 The final inequalities

The column B in Table 4 bounds the sum of the involution counts of the other maximal overgroups in Table 1. For E_7, a central fibre of size at most $q+1$ over $\mathrm{Aut}({}^2E_6(q))$ gives the stated expression. The special $E_7(2)$ overgroup is bounded by $2(2^{35}+2^{34})$, which is smaller. For E_6^ϵ, the extension of index three contributes no new involutions, and any central covering kernel is odd; apply the strengthened form of (1) and (6) over q^3. For F_4 there are at most two overgroups, each in $\mathrm{Aut}({}^3D_4(q))$. For G_2, the possible central kernel of $\mathrm{SU}_3(q).2$ has order at most three, giving $6(q^5+q^4)$. These estimates deliberately allow every indicated kernel.

G	C	B	$\lvert W\rvert$
E_8	$2q^{84}(q^{20}+q^{19})$	0	696729600
E_7	$2q^{42}(q^{12}+q^{11})$	$2(q+1)(q^{42}+q^{41})$	2903040
E_6^ϵ	$2(q^{28}+q^{27})$	$2(q^{15}+q^{12})$	51840
$F_4,\ q\geq 3$	q^{22}	$4(q^{16}+q^{15})$	1152
$G_2,\ q\geq 5$	q^5	$6(q^5+q^4)$	12
${}^3D_4$	q^{11}	0	192
${}^2F_4$	q^{11}	0	1152
${}^2G_2$	$2q^2$	0	12
${}^2B_2$	$q-1$	0	8

Table 4: Bounds charged to the two kinds of obstruction.

For E_6^ϵ we may replace D by the larger $\Phi_9(q)$, and for $E_7(2)$ by $\Phi_{18}(2)\Phi_2(2)=171$. Let L be the right side of (7), using the exact Suzuki value instead. We verify

$$L-\kappa DC-B-|W|D>0. \tag{8}$$

For the ordinary groups expand the left side at $q=q_0+z$, with $q_0=2$ for E_8,E_7,E_6^ϵ, $q_0=3$ for F_4, $q_0=5$ for G_2, and $q_0=4$ for ${}^3D_4$. For ${}^2F_4$ and ${}^2B_2$ put $q=2s^2$ and expand at $s=2+z$; for ${}^2G_2$ put $q=3s^2$ and expand at $s=3+z$. Every coefficient is nonnegative and the constant is positive in all nine cases. The supplied rational-arithmetic calculation records the full polynomials and their translated coefficients, rather than sampling field sizes.

At ${}^3D_4(2)$ and ${}^3D_4(3)$ use $\kappa=1$. The exact margins in (8) are 3648 and 17155227/2. At $F_4(2)$ the selected class has size 350859600; the two overgroups give $B\leq 4(2^{20}+2^{19})$. With $D=17$, $C=2^{22}$ and $\kappa=2$ the margin is 201942224. These calculations cover every remaining small parameter. Lemma 2, together with (3), now proves Theorem 1.

6 Attribution and verification scope

The existence result belongs to [1]. The choice of tori and the strategy of excluding maximal subgroups and involution centralizers are also borrowed from that proof. The subgroup classification, conjugacy class orders, outer-automorphism classification and uniform upper bound remain explicit external premises, cited where used. They are not new theorems of this project.

The additions here are the outer-coset criterion, the root-lattice centralizer argument including the finite checks, the flag-variety argument identifying the inverting class, the explicit normalizer charge, and the complete constant inequalities. The latter distinguish i_2 from ν_2 before passing through an extension. No asymptotic estimate supplies an unverified finite constant. The arithmetic certificates prove precisely the polynomial and lattice assertions they state; they do not prove the retained classification theorems. The flag-variety argument was supplied during review; that contribution and its separate review are recorded explicitly in the accompanying audit. The full L045 admission additionally requires its classical, alternating and sporadic components. No new conjecture resolution, Lean verification or compression measurement is claimed by this component.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, Sections 2–3.

[2] R. M. Guralnick and W. M. Kantor, *Probabilistic generation of finite simple groups*, J. Algebra 234 (2000), 743–792.

[3] W. M. Kantor and À. Seress, *Prime power graphs for groups of Lie type*, J. Algebra 247 (2002), 370–434.

[4] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), 393–464.

[5] T. C. Burness and A. R. Thomas, *On the involution fixity of exceptional groups of Lie type*, arXiv:1707.03562v2 (2018).

[6] M. Aschbacher and G. M. Seitz, *Involutions in Chevalley groups over fields of even order*, Nagoya Math. J. 63 (1976), 1–91.

[7] M. Aschbacher and G. M. Seitz, *Corrections to "Involutions in Chevalley groups over fields of even order"*, Nagoya Math. J. 72 (1978), 135–136.

Orthogonal tori and their projective orders

Human Compression project

This working note supplies the torus interface for the orthogonal part of the generation argument. It is not yet a reviewed component. It does not classify maximal overgroups or estimate the centralizers of reversing involutions. In particular it does not claim the full existence theorem.

Put $E = \mathbb{F}_{q^{2k}}$ and $E_0 = \mathbb{F}_{q^k}$, and give E, as a vector space over $\mathbb{F}_q$, the quadratic form

$$Q(v) = \mathrm{Tr}_{E_0/\mathbb{F}_q}(cv^{q^k+1}), \qquad c \in E_0^\times.$$

It is a nondegenerate form of minus type. Here is also a direct check of the type: every nonzero value of the norm has $q^k + 1$ preimages, so the number of zeros is $1 + (q^{k-1} - 1)(q^k + 1) = q^{2k-1} - (q - 1)q^{k-1}$. Nondegeneracy follows from the nondegenerate trace pairing and the polar form of the quadratic norm. Multiplication by the norm-one group is a cyclic group of isometries of order $q^k + 1$.

Lemma 1. *Suppose q is odd and z generates this norm-one group, with $q^k + 1 > 2$. Multiplication by z has nontrivial spinor norm.*

Proof. We use the determinant formula for the spinor norm in [1, Definition 3.1]; this formula and its homomorphism property are borrowed. Since -1 is not an eigenvalue of multiplication by z, that formula is the square class of $\det((1 + z)/2) = \mathrm{N}_{E/\mathbb{F}_q}(1 + z)/2^{2k}$. Choose $w \in E$ of order $2(q^k + 1)$ with $w^2 = z$. Such a choice exists because $2(q^k + 1)$ divides $q^{2k} - 1$. Then $w^{q^k} = -w^{-1}$. For $u = w + w^{-1}$ we have $u \neq 0$ and $u^{q^k} = -u$, and hence $u^2 \in E_0$ is a nonsquare there. Indeed a square root in E_0 would have to be u or $-u$. Now

$$\mathrm{N}_{E/E_0}(1 + z) = (1 + z)(1 + z^{-1}) = u^2.$$

The norm from a finite extension of finite fields sends a generator of the multiplicative group to a generator. Consequently it sends a nonsquare to a nonsquare. The displayed determinant has nonsquare class, as claimed. Multiplication by z has determinant one, so its powers lying in Ω are precisely its even powers. □

Lemma 2. *Let $q \geq 2$, $k \geq 1$, and $M > 2$ divide $q^k + 1$. If $M > q^{k/3} + 1$, then the multiplicative order of q modulo M is $2k$.*

Proof. That order is $2d$ with $d \mid k$ and k/d odd: it divides $2k$, but the power q^k has order two. If $d < k$, then $d \leq k/3$, and $q^d \equiv q^k \equiv -1 \pmod{M}$. Thus $M \leq q^d + 1$, contrary to the assumption. □

The required inequalities hold throughout the ranges below. For odd $q \geq 3$ and $k \geq 3$, $(q^k + 1)/4 > q^{k/3} + 1$: setting $y = q^{k/3} \geq 3$, this is $y^3 - 4y - 3 > 0$, which follows from its value and positive derivative at $y = 3$. For even q we use $M = q^k + 1$ directly, with $k \geq 3$.

Proposition 3. *Use the following natural orthogonal isometries and let x denote their projective images.*

1. *In minus dimension $2m$, $m \geq 4$, put $N = q^m + 1$. Use a primitive norm-one multiplier g when q is even and g^2 when q is odd. Its projective order is N in even characteristic and $N/\gcd(4, N)$ in odd characteristic.*

2. *In odd dimension $2m + 1$, q odd and $m \geq 3$, use g^2 on a negative $2m$-space and the identity on its orthogonal line. Its order is $(q^m + 1)/2$.*

3. *In plus dimension $2m$, $m \geq 4$, put $A = q^{m-1} + 1$ and $B = q + 1$. On negative spaces of dimensions $2m - 2$ and 2, use any primitive norm-one multipliers g, h. For even q the projective order is AB when m is odd and A when m is even. For odd q it is*

$$\begin{cases} AB/\gcd(4, q^m - 1), & m \text{ odd}, \\ A/2, & m \text{ even}. \end{cases}$$

All these isometries belong to the indicated omega groups.

Proof. In odd characteristic, Lemma 1 gives omega membership in the first two cases and cancellation of the two nontrivial spinor norms in the third. In even characteristic the elements have odd order, so their images under the Dickson homomorphism to a group of order two vanish. These homomorphisms and the determinant condition defining omega are as in [1, Section 3].

The order of g^2 in the first case is $N/2$. The scalar $-I$ belongs to this cyclic subgroup if and only if $4 \mid N$, which gives exactly the additional divisor stated. In odd dimension the fixed line prevents any power other than the identity from being scalar.

For the third case, if m is odd, the Euclidean algorithm gives $\gcd(A, B) = \gcd(2, q+1)$. For odd q, A has exactly one factor of two. The pair (g, h) has order $AB/2$; its half-order power is $(-I, -I)$ exactly when B also has exactly one factor of two, namely when $q \equiv 1 \pmod 4$. If m is even, $B \mid A$ and A/B is odd. The pair has order A, and for odd q its $A/2$ power is $(-I, -I)$. There is no nontrivial scalar isometry in even characteristic. These observations prove all the projective order formulas. □

Proposition 4. *In the natural projective conformal orthogonal group, the centralizers of the elements in Proposition 3 are tori of orders N, N, AB, respectively. They lie in the connected adjoint component. There is no involutory reversing automorphism represented on this natural module by a semilinear map with nontrivial ordinary field part. The assertion does not address the additional triality automorphisms in plus dimension eight.*

Proof. On the large negative space let α be the chosen multiplier, and write M for its order after identifying scalar signs. In the minus case $M = N/\gcd(4, N)$ for odd q; in the plus case it is $A/2$ for odd q; in even characteristic it is the full norm-one order. In the odd dimensional case we may use the smaller value $N/\gcd(4, N)$ too. Lemma 2 and the inequalities following it show that $\mathrm{ord}_M(q) = 2k$, where $k = m$ or $m - 1$. In particular the large block is irreducible, and is distinguished from the other blocks by its dimension.

If a conformal map Y centralizes the projective image of the chosen isometry X, then $YXY^{-1} = \lambda X$. Comparing the preserved forms gives $\lambda^2 = 1$. On the large block the eigenvalues of the two similar matrices therefore give $\alpha^{q^j} = \lambda\alpha$ for some $0 \leq j < 2k$. Modulo scalar signs this says $q^j \equiv 1 \pmod M$, so $j = 0$ and $\lambda = 1$. Hence Y centralizes X as a linear map.

Its restriction to the large block belongs to the field algebra $E = \mathbb{F}_q[X]$, and the conformal condition on multiplication by z is $\mathrm{N}_{E/E_0}(z) \in \mathbb{F}_q^\times$. This follows by substituting into Q and using the nondegenerate trace pairing. There are $(q-1)(q^k+1)$ such multipliers. In minus type division by the $q - 1$ scalar multipliers gives N. In plus type the two block multipliers must have the same norm in $\mathbb{F}_q^\times$. The number of pairs is $(q - 1)AB$, so their projective group has order AB. The small block is irreducible because its primitive multiplier has order $q + 1 > 2$;

this includes $q = 2$. In odd dimension the line multiplier a forces the large-block norm to be a^2. Dividing both multipliers by a leaves precisely the norm-one group, of order N. These multiplicative centralizers are connected tori over the algebraic closure and consequently belong to the connected adjoint component.

Finally let $q = p^f$. An involutory automorphism with a nontrivial field projection has f even and projection $p^{f/2}$. If it reverses x, comparison of the eigenvalues on the large block, again allowing the scalar sign, implies

$$p^{f/2+fj} \equiv -1 \pmod{M}$$

for some j (the sign introduced by a scalar is already quotiented out in M). Squaring gives $q^{2j+1} \equiv 1 \pmod{M}$. This contradicts $\mathrm{ord}_M(q) = 2k$. Natural graph automorphisms are represented by linear conformal maps and do not alter this argument; exceptional triality still requires a separate argument. □

The distinction between a cyclic subgroup and its centralizer matters. For example in plus type with even m, x has order $A/2$ for odd q but its adjoint torus has order AB. In minus dimension ten over $\mathbb{F}_3$, the projective Singer image has order 61, whereas the natural omega Singer subgroup has order 122. The latter convention agrees with [2, Table 1]. We retain these scalar losses explicitly in the later counting argument.

References

[1] S. H. Murray and C. M. Roney-Dougal, *The spinor norm and homomorphism algorithms for classical groups*, preprint dated 2 May 2008, Definition 3.1 and Lemma 3.2.

[2] D. Bubboloni, M. S. Lucido and T. Weigel, *2-Coverings of classical groups*, arXiv:1102.0660 (2011).

Bounds for orthogonal fixed groups in odd characteristic

Human Compression project

This component supplies fixed-group bounds and two subspace counts for the orthogonal generation argument. Throughout q is odd and $\nu_2(H) = |\{h \in H : h^2 = 1\}|$, including the identity. We write $\mathrm{PCO}(V)$ for the full conformal orthogonal group modulo the scalar group over its ground field. This avoids confusion with the image of the isometry group, which can be a proper subgroup.

We retain the standard finite classical group orders, the faithful action of projective conformal groups on their derived simple groups, and Lawther–Liebeck–Seitz, Proposition 1.3 [1]. The latter gives $\mathrm{i}_2(\mathrm{Aut}\, S) < 2(Q^{N+r} + Q^{N+r-1})$ over $\mathbb{F}_Q$. The actual torus centralizers and projective eigenvalue degrees are acquired from the separately reviewed orthogonal torus component [3].

Lemma 1. *For $d \geq 2$ and a nondegenerate d-space over $\mathbb{F}_q$, put $e_d = \lfloor d^2/4 \rfloor$. Then*

$$\nu_2(\mathrm{PCO}_d(q)) \leq 3q^{e_d}, \qquad |\mathrm{PCO}_d(q)| < 3q^{d(d-1)/2}.$$

Proof. For $d \geq 5$, the derived projective group is simple, including the identifications in dimensions five and six. Its exponent $N + r$ is e_d. The faithful projective action and the retained bound give $\nu_2(\mathrm{PCO}_d(q)) < 2q^{e_d} + 2q^{e_d-1} + 1 \leq 3q^{e_d}$.

In dimension two the group is dihedral of order $2(q - \epsilon)$; its number of elements of square one is $q - \epsilon + 2 \leq q + 3$. In dimension three it is $\mathrm{PGL}_2(q)$, with q^2 nonidentity involutions. In dimension four the two models, already verified in the linear/unitary component [4], are

$$\mathrm{PGL}_2(q) \wr C_2, \qquad \mathrm{PGL}_2(q^2) \rtimes C_2.$$

The respective values of ν_2 are $q^4 + q^3 + 2q^2 - q + 1$ and $q^4 + q^3 + q + 1$, both at most $3q^4$. This proves the first assertion in every dimension needed here.

For odd d, the projective conformal group has the order of $\mathrm{SO}_d(q)$, less than $q^{d(d-1)/2}$. For even $d = 2s$, it has the order of the full isometry group,

$$2q^{s(s-1)}(q^s - \epsilon)\prod_{j=1}^{s-1}(q^{2j} - 1) \leq 2(1 + q^{-s})q^{d(d-1)/2} \leq \tfrac{8}{3}q^{d(d-1)/2}.$$

The ordinary order formulas also include dimension two. □

Proposition 2. *Let V have dimension $2m$, $m \geq 3$. Suppose a conformal map a has square cI, with multiplier c, and its two eigenspaces over the algebraic closure both have dimension m. Then*

$$\nu_2(C_{\mathrm{PCO}(V)}([a])) \leq 24q^{2\lfloor m^2/4 \rfloor}.$$

If c is a square and the two eigenspaces instead have unequal dimensions $r, s \geq 2$, then

$$\nu_2(C_{\mathrm{PCO}(V)}([a])) \leq 18q^{e_r+e_s}.$$

Proof. First suppose c is a square. Rescale a to have square one. Its two eigenspaces are nondegenerate and orthogonal. A map centralizing $[a]$ sends a to a or $-a$, since it preserves the multiplier. It therefore preserves the two eigenspaces, or interchanges them. Projecting its restrictions to their projective conformal groups has kernel of order at most two: a scalar on each block must have the same square, because the two multipliers agree, and a common scalar is already factored out.

When an interchange is possible the two forms are similar. Writing H for either projective conformal group, the image is a subgroup of $H \wr C_2$. There are $\nu_2(H)^2$ elements of square one in the base, and exactly $|H|$ in the interchange coset, since its two coordinates must be mutual inverses. A group extension with kernel of order at most two has at most twice as many elements of square one as its image. Lemma 1 thus gives

$$\nu_2(C) \le 2\big(9q^{2e_m} + 3q^{m(m-1)/2}\big) \le 24q^{2e_m}.$$

If the forms are not similar, omit the interchange coset and use the product bound. Unequal dimensions also forbid interchange, giving $2(3q^{e_r})(3q^{e_s})$ as stated.

Now suppose c is a nonsquare. The algebra $K = \mathbb{F}_q[a]$ is $\mathbb{F}_{q^2}$. The map a is self-adjoint for the underlying bilinear form, because $a^* = ca^{-1} = a$. The nondegenerate trace pairing then writes that form uniquely as the trace of a nondegenerate symmetric K-bilinear form on the m-dimensional K-space V. A projective centralizer acts either linearly or by the nontrivial field automorphism of K; the induced projective group embeds in the automorphism group of the corresponding derived orthogonal group over K. Its kernel has order at most two. Indeed a scalar $z \in K^\times$ induces an allowed similitude over $\mathbb{F}_q$ only if $z^2 \in \mathbb{F}_q^\times$, and there are exactly two such cosets modulo $\mathbb{F}_q^\times$.

For $m \ge 5$, simplicity and the retained involution bound give $\nu_2(C) \le 6q^{2e_m}$. The same argument applies when $m = 3$, using $\mathrm{P}\,\Omega_3(K) \simeq \mathrm{PSL}_2(K)$. For $m = 4$ of minus type it uses $\mathrm{P}\,\Omega_4^-(K) \simeq \mathrm{PSL}_2(q^4)$, and gives $6q^8$.

The plus case in dimension four needs its two simple factors. Put $Q = q^2$ and $A = \mathrm{PSL}_2(Q)$; the derived projective group is $A \times A$. Its full automorphism group is $\mathrm{Aut}(A) \wr C_2$. We have $\nu_2(\mathrm{Aut}\,A) \le 3Q^2$ and $|\,\mathrm{Aut}\,A| = f_Q Q(Q^2-1) < Q^4$, where $Q = p^{f_Q}$ and $f_Q \le Q$. Thus

$$\nu_2(C) \le 2\big(9Q^4 + Q^4\big) = 20q^8.$$

These bounds all imply the stated constant 24. □

Proposition 3. *For a natural involutory reversing automorphism of any of the chosen orthogonal elements over odd $\mathbb{F}_q$, its fixed subgroup in the simple group has at most*

$$\begin{cases} 24q^{2\lfloor m^2/4 \rfloor}, & \dim V = 2m, \\ 18q^{m(m+1)/2}, & \dim V = 2m+1 \end{cases}$$

elements of square one. This assertion does not count triality reversers which lack a representative on the natural module.

Proof. Let X be the natural representative of the chosen torus element. Projective inversion initially gives $aXa^{-1} = \lambda X^{-1}$, with $\lambda^2 = 1$. On the large block, the acquired degree lemma forces the field exponent of a to be k, where the block has dimension $2k$. Since the eigenvalue α satisfies $\alpha^{q^k} = \alpha^{-1}$ exactly, the scalar is $\lambda = 1$. Consequently the restriction is $z\,\mathrm{Frob}^k$. Its square and conformal multiplier are both $\mathrm{N}_{\mathbb{F}_{q^{2k}}/\mathbb{F}_{q^k}}(z)$, which lies in $\mathbb{F}_q^\times$. The same calculation on the negative two-plane gives the same scalar; on a fixed line its multiplier is the square of its line scalar. Thus $a^2 = cI$, with multiplier c.

If c is a square, on every negative $2k$-block a scalar multiple of a is a semilinear involution for the quadratic extension over $\mathbb{F}_{q^k}$. Its fixed and negative spaces both have dimension k.

This follows also by solving $zv^{q^k} = \pm v$: norm one is precisely the solvability condition, and each solution space is one-dimensional over $\mathbb{F}_{q^k}$. The dimensions are therefore m, m in even dimension and $m, m+1$ in odd dimension. In the latter case the line forces c to be a square. Apply Proposition 2; taking a subgroup cannot increase its number of elements of square one. □

Proposition 4. *For odd q the following strict lower bounds hold:*

$$\mathrm{i}_2(P\Omega_8^+(q)) > q^{16}/4, \qquad \mathrm{i}_2(P\Omega_8^-(q)) > q^{16}/2.$$

Proof. In either space let W run over the nondegenerate four-dimensional subspaces of plus type. Put $t_W = -I$ on W and $t_W = I$ on its orthogonal complement. Its determinant is one and its spinor norm is the discriminant of the four-dimensional plus form, a square. Thus $t_W \in \Omega(V)$. We use here the standard spinor norm of $-I$ on a nondegenerate subspace, as in [2, Definition 3.1]. Witt's extension theorem makes the full isometry group transitive on these subspaces, with stabilizer $O(W) \times O(W^\perp)$.

In plus dimension eight the complement also has plus type. The projective images of t_W and $t_{W'}$ agree exactly when $W' = W$ or $W' = W^\perp$: the scalar identifying two isometries must be 1 or -1. No nondegenerate W equals its complement. Hence the resulting distinct involutions number

$$\frac{|O_8^+(q)|}{2|O_4^+(q)|^2} = \frac{q^8(q^2+1)^2(q^4+q^2+1)}{4} > \frac{q^{16}}{4}.$$

In minus dimension eight the complement has minus type, while $-I$ has nonsquare spinor norm and does not lie in $\Omega(V)$. There is no projective identification between the displayed involutions. Their number is

$$\frac{|O_8^-(q)|}{|O_4^+(q)|\,|O_4^-(q)|} = \frac{q^8(q^4+1)(q^4+q^2+1)}{2} > \frac{q^{16}}{2}.$$

The count is of actual subspaces, so it does not require transitivity of the smaller omega group on this set. □

Scope. The constants deliberately include the identity, possible projective order-four lifts, a nonsplit quadratic-field form, and interchange cosets. No central quotient is estimated by counting only involutions in its covering group. The separate covering and total-involution estimates remain necessary for a generation conclusion.

References

[1] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific Journal of Mathematics 205 (2002), Proposition 1.3. The full automorphism bound, including the identity adjustment here, is retained, not reproved.

[2] S. H. Murray and C. M. Roney-Dougal, *The spinor norm and homomorphism algorithms for classical groups*, preprint dated 2 May 2008, Definition 3.1 and Lemma 3.2.

[3] Human Compression project, *Orthogonal tori and their projective orders*, accepted component with independent review.

[4] Human Compression project, *Linear and unitary groups*, accepted component; the precise dimension-four projective conformal models and their counts are acquired.

Finite kernels of full-rank orthogonal inverters in characteristic two

Human Compression project

Let q be even, let $m \geq 2$, let V be a nonsingular quadratic space of dimension $2m$, and let $t \in O(V)$ have $(t-I)^2 = 0$ and rank m. Write B for the polar form and assume $B(v, tv)$ is not identically zero. These are the full-rank types c_m for even m, and b_m for odd m in Aschbacher–Seitz Sections7–8. The result below also covers $t \notin \Omega(V)$: the centralizer is always taken in $\Omega(V)$. Put $\nu_2(H) = |\{h : h^2 = 1\}|$.

Theorem 1. *There are surjective homomorphisms with the indicated finite kernel orders:*

	image of $C_{\Omega(V)}(t)$	*kernel order*
m odd	$\mathrm{Sp}_{m-1}(q)$	$q^{(m^2-m)/2}$
m even	$\mathrm{Sp}_{m-2}(q)$	$q^{(m^2+m-2)/2}$.

Here Sp_0 *is trivial. Consequently the centralizer's* ν_2 *is at most the kernel order times* ν_2 *of the displayed symplectic group. In particular, when* $m = 4$*,*

$$\nu_2(C_{\Omega(V)}(t)) \leq q^9 q^2 = q^{11}.$$

No order is inferred from an algebraic dimension.

1 A direct finite fibre calculation

The nonalternating full-rank symplectic normal form from AS7.6–7.7 permits coordinates $x, y \in \mathbb{F}_q^m$ in which

$$t(x, y) = (x + y, y), \qquad B((x, y), (x', y')) = x^t y' + y^t x'.$$

Equivalently, the nonsingular nonalternating symmetric matrix defining the square-zero part is congruent to I_m over the perfect field. Let $s = (1, \ldots, 1)^t$. A quadratic form with this polar form differs from $x^t y$ by squares of linear forms. Its invariance under t forces its x-part to be $(s^t x)^2$. Thus, for some column b, it is exactly

$$Q(x, y) = (s^t x)^2 + x^t y + b^t y^{[2]}, \tag{1}$$

where [2] denotes entrywise squaring.

Block commutation and the symplectic equation give every symplectic centralizer element uniquely as

$$g_{A,D}(x, y) = (Ax + DAy, Ay), \qquad AA^t = I_m, \quad D = D^t.$$

Put $J_m = \{A : AA^t = I_m\}$. These matrices fix s, since $v^t v = (s^t v)^2$ and squaring is injective. Substituting $w = Ay$ in (1) proves that $g_{A,D}$ is orthogonal precisely when

$$F(D) := \mathrm{diag}(D) + (Ds)^{[2]} = b + A^{[2]} b. \tag{2}$$

This is an equation of additive groups, not an asserted $\mathbb{F}_q$-linear map.

We count its fibres. Put $d = m(m-1)/2$. If c_i is the sum of the off-diagonal entries in row i, then $\sum c_i = 0$. The map from the d off-diagonal entries to (c_i) is onto that hyperplane: entries in the last column prescribe its first $m-1$ coordinates. Every fibre has q^{d-m+1} elements. Writing $a_i = D_{ii}$, equation $F(D) = 0$ becomes

$$a_i^2 + a_i = c_i^2.$$

Over $\mathbb{F}_q$, the map $a \mapsto a^2 + a$ has kernel of order two and image $\ker \mathrm{Tr}_{q/2}$. Since $\mathrm{Tr}(c_i^2) = \mathrm{Tr}(c_i)$, there are $(q/2)^{m-1}$ possible row-sum vectors, then q^{d-m+1} off-diagonal choices, and 2^m diagonal choices. Therefore

$$|\ker F| = 2q^d. \tag{3}$$

Also, writing $a = \sum_i D_{ii}$, we have $\sum_i F(D)_i = a + a^2$. It follows that the image of F lies in $\{c : \mathrm{Tr}_{q/2}(\sum c_i) = 0\}$. Both sets have size $q^m/2$ by (3), so they are equal. Since $s^t A^{[2]} = s^t$, the right side of (2) has coordinate sum zero. Thus every $A \in J_m$ lifts, and every fibre in $C_{O(V)}(t) \to J_m$ has exactly $2q^d$ elements.

The matrix $g_{I,E_{11}}$ is in this kernel: $F(E_{11}) = 0$. It has rank one after subtraction of the identity, so its Dickson invariant is one. We retain the standard Dickson homomorphism $O(V) \to C_2$, $g \mapsto \mathrm{rank}(g - I) \bmod 2$, whose kernel is $\Omega(V)$. Multiplication by this kernel element toggles that invariant without changing A. Therefore every fibre has exactly q^d elements in $\Omega(V)$. We have proved the surjection

$$C_{\Omega(V)}(t) \longrightarrow J_m, \qquad |\ker| = q^d. \tag{4}$$

This argument pays the full orthogonal component and its index-two intersection; no connectedness or splitting assumption replaces the count.

2 The smaller symplectic image

For odd m, $s^t s = 1$, and $s^\perp$ is a nonsingular alternating space. Every member of J_m fixes s, and conversely every symplectic isometry of $s^\perp$ extends by fixing s. Hence $J_m = \mathrm{Sp}_{m-1}(q)$.

For even m, $s^t s = 0$. Choose e with $s^t e = 1$, whence $e^t e = 1$, and let $H = \langle s, e\rangle^\perp$. The form on H is nonsingular alternating of dimension $m-2$. The action on $s^\perp/\langle s\rangle$ gives a surjection $J_m \to \mathrm{Sp}(H)$, split by fixing s, e. Every element of its kernel, and no other one, has the form

$$s \mapsto s, \qquad h \mapsto h + (h^t u)s \ (h \in H), \qquad e \mapsto e + u + as, \qquad u \in H,\ a \in \mathbb{F}_q.$$

These maps preserve every displayed pairing. Conversely, fixing the quotient forces the stated shape, and the pairing with e determines its linear functional. Thus the kernel has exactly q^{m-1} elements. Combining with (4) proves the exact kernel orders in Theorem 1. These kernels have orders which are powers of two. They are therefore unipotent matrix groups in characteristic two; their orders above are finite counts, including the orthogonal component factor.

For any surjection $C \to L$ with finite kernel K, an element of square one maps to an element of square one. There are at most $|K|$ such elements above each image, so $\nu_2(C) \le |K|\nu_2(L)$. In characteristic two, $\mathrm{Sp}_2(q) = \mathrm{SL}_2(q)$ has exactly $q^2 - 1$ nonidentity involutions: each is a rank-one unipotent, with one eigenline, and for each of the $q+1$ lines there are $q-1$ nontrivial transvections. Thus $\nu_2(\mathrm{Sp}_2(q)) = q^2$, which proves the stated q^{11} bound.

3 Application to the natural torus inverters

In the acquired orthogonal torus models, a natural projective involutory inverter in characteristic two has an isometric lift of square one. Indeed if a conformal lift has square aI, its multiplier is a;

rescaling by the unique square root of a^{-1} achieves both conditions. Projective inversion becomes literal inversion, since the only scalar isometry in characteristic two is one.

On a negative $2k$-block write $E = \mathbb{F}_{q^{2k}}$, $\bar{v} = v^{q^k}$ and $Q(v) = \mathrm{Tr}_{q^k/q}(c_0 v\bar{v})$, $c_0 \neq 0$. The acquired torus degree argument distinguishes this irreducible block from the others. If its multiplier is α, then $\alpha^{-1} = \bar{\alpha}$, and an inverting linear map satisfies $t(\alpha v) = \bar{\alpha} t(v)$. Since $\mathbb{F}_q[\alpha] = E$, it is exactly $v \mapsto c\bar{v}$. Its square is one precisely when $c\bar{c} = 1$. The small negative plane in the plus model has the same argument, since its primitive multiplier has order $q + 1 > 2$. Hilbert90, or the cyclic multiplicative group directly, conjugates it linearly to $v \mapsto \bar{v}$, so its square-zero part has rank k. Moreover its polar form satisfies

$$B(v, c\bar{v}) = \mathrm{Tr}_{E/\mathbb{F}_q}(c_0 \bar{c}\, v^2),$$

which is not identically zero by the nondegenerate trace pairing and bijectivity of squaring. The minus model therefore has rank m and is nonalternating. In the plus model the two negative blocks give ranks $m - 1$ and one; restriction to the large block already makes the full form nonalternating. They therefore have precisely the types c_m (m even) and b_m (m odd), and Theorem 1 applies for either orthogonal sign. No flag-dimension test is required to eliminate a_m.

Source and scope. AS1976 Sections7–8 supplies the symplectic normal form and class labels; the complete relevant passages and the 1978 corrections were read. The correction to AS8.8(i) concerns the last column of P; here the middle space has dimension zero, and the direct equation (2) independently pays all finite fibres. The Dickson convention is retained from Murray–Roney-Dougal2008, Definition3.1(2), Lemma3.2 and Definition3.3. The coordinate fibre count and its use here are new derivations by publication_recovery; root requested this interface and proposed its expected exponents before this proof. These arguments do not pay triality, maximal-overgroup classification, odd-characteristic fixed groups or the whole L045 claim.

References

[1] M. Aschbacher and G. M. Seitz, *Involutions in Chevalley groups over fields of even order*, Nagoya Math. J. 63 (1976), 1–91, Sections 7–8; corrections, Nagoya Math. J. 72 (1978), 135–136.

[2] S. H. Murray and C. M. Roney-Dougal, *The spinor norm and homomorphism algorithms for classical groups*, preprint dated 2 May 2008, Section 3.

Maximal overgroups of a full negative $6+2$ torus element

L045 recovery: a classification interface

Let $q = p^f \geq 4$, $d = (2, q-1)$, $A = q^3+1$, and $B = q+1$. Let $V = V_6^- \perp V_2^-$ over $\mathbb{F}_q$, and let $X = g \perp h$ where g and h generate the norm-one tori of orders A and B on the two summands. Thus $X \in \Omega_8^+(q)$ and its image $x \in G = P\Omega_8^+(q)$ has order A/d, by the acquired torus proof. The primitive elements g, h are arbitrary; no relation between their choices is assumed.

Theorem 1. *Every maximal subgroup of G containing x is one of the following:*

1. *the unique stabilizer D of $V_6^- \perp V_2^-$;*
2. *the normalizer of a unitary structure on a four-dimensional $\mathbb{F}_{q^2}$-space; there are at most two such subgroups containing x;*
3. *a subgroup of type B_3: $\Omega_7(q)$ for odd q, or $\mathrm{Sp}_6(q)$ for even q, in one of the three triality actions.*

The natural-action B_3 subgroups do not contain x. Also $N_G(\langle x \rangle) \leq D$, so a separate cyclic-normalizer term is not missing from this cover.

The only classification premise is the complete published maximal-subgroup classification in Bray–Holt–Roney-Dougal (BHRD), Main Theorem 2.1.1, Section 8.1 and Table 8.50 [1]. The table is explicitly attributed there to Kleidman's classification of these orthogonal groups. We use the table for the simple group G, not for arbitrary almost simple extensions. Its novelty rows therefore give no additional maximal subgroup of G. All eliminations for our particular elements are proved below. The complete row-by-row routing is retained separately.

Lemma 2. *There is a prime $r \mid |x|$ with $\mathrm{ord}_r(p) = 6f$. Every such prime satisfies $r \geq 6f+1$ and $\mathrm{ord}_r(q) = 6$. Moreover the entire primitive part $\Phi_{6f}^*(p)$ divides $|x|$.*

Proof. The primitive-divisor theorem applied to $p^{6f}-1$ has no exception here: the exceptional pair $(p, 6f) = (2, 6)$ would mean $q = 2$. All primitive prime powers divide $p^{3f}+1 = A$, and none divides 2. This proves the assertions after passing to the projective quotient of order $d \leq 2$. We retain the classical primitive-divisor theorem as a named number-theoretic input. Its exact statement and the primitive-part convention are also recalled in the acquired Bamberg–Penttila preprint [2, pp.2–3]. □

Proof of the type assertion in Theorem 1. Apply the complete BHRD table to the preimage of a maximal subgroup in $\Omega_8^+(q)$. Central factors of order at most two do not affect the primitive divisor. The parabolic rows have Levi factors of types $\mathrm{GL}_4(q)$, $\mathrm{GL}_2(q)\Omega_4^+(q)$ or $\mathrm{GL}_3(q)$; their orders have no factor r. The same test excludes the positive $2+6$ and SL_4 rows, the $3+5$ and $\mathrm{Sp}_2 \circ \mathrm{Sp}_4$ rows, the four-plane decompositions, the two four-space decompositions, and $\Omega_4^+(q^2)$. Their non-characteristic order factors divide products of q^j-1 for $j \leq 4$, together with powers of 2 and 3. The additional dihedral novelty row has factors q^2+1 and powers of two, and is excluded by the same argument. The negative $2+6$ and unitary four-space rows survive.

The subfield rows are excluded using the *absolute* order $6f$. For $q = q_0^s$, $q_0 = p^a$, $s \geq 2$, the factors in $\Omega_8^+(q_0)$ have exponents at most $6a < 6f$. For the two $\Omega_8^-(q_0)$ actions with $q = q_0^2$, the exponents are at most $8a < 12a = 6f$. For ${}^3D_4(q_0)$ with $q = q_0^3$, use

$$|{}^3D_4(q_0)| = q_0^{12}(q_0^8+q_0^4+1)(q_0^6-1)(q_0^2-1), \qquad q_0^8+q_0^4+1 \mid q_0^{12}-1.$$

Here $12a < 18a = 6f$. All displayed extension factors in these table rows have order a power of two. The $L_3(q).3$ row has factors of exponents at most three and an extra factor three, so is excluded too.

Consider the $U_3(q).3$ row, which occurs only when $q \equiv 2 \pmod 3$. In its projective group, $y = x^3$ lies in $\mathrm{PSU}_3(q)$ and has order $A/(3d)$. An r-element in $\mathrm{PSU}_3(q)$ acts irreducibly on its natural three-space over $\mathbb{F}_{q^2}$, since $\mathrm{ord}_r(q^2) = 3$. Its centralizer is the norm-one Singer torus modulo scalars, of order $(q^2-q+1)/3$. This follows directly by identifying the commuting algebra with $\mathbb{F}_{q^6}$: the unitary condition is norm one to $\mathbb{F}_{q^3}$, the determinant-one condition leaves order q^2-q+1, and the scalar center has order three. Projective centralization gives no extra elements: lift the r-element with order r; a commuting lift can differ by a scalar of order three, which its rth power forces to be one. Since y centralizes its own r-subgroup, its order must divide this torus order. This is impossible, their ratio being $B/d > 1$.

The remaining bounded-order rows are handled as follows. The monomial and extraspecial rows have a normal 2-group with quotient contained in S_8 (or $L_3(2)$), and require $q = p$ odd. Their primitive part is therefore at most seven: a primitive divisor is at least seven, and seven occurs only once in those quotient orders. The same assertion holds for the row $2 \cdot \Omega_8^+(2)$, whose prime divisors are $2, 3, 5, 7$, with only one factor seven. But

$$\Phi_6^*(p) = \frac{p^2 - p + 1}{(3, p+1)} > 7 \qquad (p \geq 7).$$

Only $q = 5$ remains in our range. Now $|x| = 63$. An odd-order element injects through a 2-group kernel; S_8 and $L_3(2)$ have no element of order 63. Neither does $\Omega_8^+(2)$, as proved in Lemma 3 below. The exceptional rows $2 \cdot \mathrm{Sz}(8)$ and the two central extensions of A_{10} occur only at $q = 5$. The first group has order prime to three; the second ones cannot have order 63, since a permutation of that order requires disjoint cycles of lengths nine and seven. The A_9 row has $q = 2$ and is outside the hypothesis. The G_2 row is marked as a novelty and is not maximal in the simple group G. Thus only the three stated types remain. A natural B_3 subgroup fixes a nonsingular line, whereas X has no eigenvalue in $\mathbb{F}_q$. □

Lemma 3. *No element of the orthogonal group $O_8^+(2)$ has order* 63.

Proof. Such an element would be semisimple. An eigenvalue whose order is divisible by 63 has degree at least six, and an eigenvalue of order 63 has degree six. Its irreducible polynomial is not self-reciprocal: no power of 2 is -1 modulo 63. A second, distinct dual constituent would require twelve dimensions. If no eigenvalue has order divisible by 63, separate constituents must supply the factors nine and seven. Order nine requires six dimensions, since $\mathrm{ord}_9(2) = 6$. A constituent supplying the factor seven costs at least six dimensions: if its degree is three, its eigenvalue has order seven and a distinct dual constituent is required; otherwise its degree is at least six (this includes eigenvalue order 21). The two contributions are distinct, since otherwise an eigenvalue would have order divisible by 63. Again at least twelve dimensions are necessary. □

Proof of the multiplicities and normalizer assertion. Both X and X^2 have exactly two distinct irreducible constituents, of dimensions six and two. Indeed g^2 has order $A/(2, A)$, which exceeds $q^2 - 1$ and does not divide $q^3 - 1$; its degree is therefore six. The order of h^2 is $B/(2, B) > 2$ for $q \geq 4$, so its degree is two. Thus the only proper nonzero X-invariant subspaces are V_6 and V_2, and the decomposition stabilizer D is unique.

Let $E \cong \mathbb{F}_{q^2}$ be a quadratic scalar-field structure whose unitary normalizer contains X. Conjugation by X induces an automorphism of E of order at most two, so X^2 commutes with E. Consequently

$$E \subseteq C_{\mathrm{End}_{\mathbb{F}_q}(V)}(X^2) = \mathbb{F}_{q^6} \oplus \mathbb{F}_{q^2}.$$

This algebra is commutative and contains X, so X in fact centralizes E. Each coordinate projection of E is an embedding. Up to a common Frobenius automorphism of E, there are exactly

two choices, determined by the relative Frobenius on the two factors. A field structure and the quadratic form determine its full unitary extension normalizer; hence at most two of the unitary maximal subgroups can contain x.

Finally a lift of an element normalizing $\langle x\rangle$ conjugates X to ϵX^j, where $\epsilon \in \{1,-1\}$ and $(j, A/d) = 1$. The two constituents of X^j still have degrees six and two: $(j, A) \le d$, so their orders lose at most a factor two, and multiplication by ϵ does not change their degrees. The preceding degree argument applies. Their unique dimensions force the lift to preserve both summands. Therefore $N_G(\langle x\rangle) \le D$. □

Scope and attribution. This is an application of an explicitly retained complete subgroup classification, not a new classification. It corrects the missing arbitrary-element interface in the use of Leemans–Liebeck Table 4; it does not infer that that source's multiplicities or projective torus orders are correct. The triality analysis deciding which B_3 subgroups contain a particular x, the inverter bounds and the final generating-pair inequality are separate components. No whole-original admission, measurement or new question is asserted here. The torus construction and the surrounding orthogonal program were authored by the root agent; this cover specialization and its two-field-structure argument were authored by publication_recovery. The parent independently obtained the relative-field orientation argument in a separate orthogonal manuscript. The peer f003_resume supplied a supporting elementary order-21/63 note and arithmetic checks; these are retained as a contribution, not as an independent review of this component.

References

[1] J. N. Bray, D. F. Holt and C. M. Roney-Dougal, *The Maximal Subgroups of the Low-Dimensional Finite Classical Groups*, LMS Lecture Note Series 407, Cambridge University Press, 2013. Main Theorem 2.1.1, Section 8.1, Table 8.50 (printed pp.55,373–374,402–404). Complete published book acquired from the National Academic Digital Library; the classification is retained, not recomputed.

[2] J. Bamberg and T. Penttila, *Overgroups of cyclic Sylow subgroups of linear groups*, Comm. Algebra 36 (2008), 2503–2543; complete 27-page author-hosted preprint acquired from Bamberg's Ghent preprint page, with its old-draft qualification and current author errata explicitly retained. Only its elementary primitive-part convention is used above; the complete maximal cover comes from BHRD.

Counting overgroups of an orthogonal torus element

Human Compression project

This working proof uses the previously reviewed orthogonal tori and fixed-group estimates. The maximal subgroup classifications cited below are retained inputs, not results newly proved here. The plus eight-dimensional case has a separate, still unpaid classification interface; the final paragraphs for that case are conditional on its stated cover.

Write $\nu(H) = |\{h \in H : h^2 = 1\}|$, so that the identity is included. For any homomorphism of finite groups with kernel K,

$$\nu(H) \le |K|\nu(H/K). \tag{1}$$

We use the accepted criterion that an element $x \in G$ has a generating involution y with no involutory automorphism simultaneously inverting x and fixing y, provided

$$i_2(G) > \sum_{\alpha^2=1,\ x^\alpha = x^{-1}} i_2(C_G(\alpha)) + |N_G(\langle x\rangle)| + \sum_M i_2(M). \tag{2}$$

Here the last sum is over the actual maximal subgroups containing x. Replacing these involution counts by ν only enlarges the right side. The normalizer is charged even when already contained in that union.

1 The chosen elements and the classification input

On a negative space of dimension $2k$ identify the underlying space with $E = \mathbb{F}_{q^{2k}}$ and use

$$Q(v) = \mathrm{Tr}_{\mathbb{F}_{q^k}/\mathbb{F}_q}(cv^{q^k+1}), \qquad c \neq 0.$$

Let g be a primitive norm-one multiplier. The accepted torus proof establishes its spinor norm, exact projective order, centralizer and eigenvalue degree. Use the following lifts X of x:

1. $G = \Omega_{2m+1}(q)$, q odd, $m \ge 3$: g^2 on the negative $2m$-space and the identity on its orthogonal line.
2. $G = P\Omega^-_{2m}(q)$, $m \ge 4$: g for even q, and g^2 for odd q.
3. $G = P\Omega^+_{2m}(q)$, $m \ge 4$: $g \oplus h$, with arbitrary primitive multipliers on negative spaces of dimensions $2m-2$ and two.

The connected projective conformal centralizer has respective orders

$$|T| = q^m + 1, \quad q^m + 1, \quad (q^{m-1}+1)(q+1). \tag{3}$$

We retain the following complete overgroup types from [1, Lemmas 6.1,7.3,8.1]. The statements there concern the corresponding subgroups of Ω; lifting a projective maximal subgroup through the centre gives exactly their applicable setting. In odd dimension the only type is $\Omega^-_{2m}(q).2$, except for $(m,q) = (3,3)$, which is already in the separately acquired finite list. In minus dimension the types are the extension field normalizers

$$\Omega^-_{2m/r}(q^r).r, \qquad r \mid m \text{ prime}, \quad 2m/r \ge 4,$$

and a unitary normalizer of type $\mathrm{GU}_m(q).2$ when m is odd. In plus dimension $2m \ge 10$ they are the stabilizer of the negative $(2m-2)+2$ decomposition and, for even m, a normalizer of type $\mathrm{GU}_m(q).2$. Lemma 8.1 uses primitive components of precisely the orders used here. We do not replace them by their squares.

2 Counting actual subgroups

A list of conjugacy types is insufficient for (2). We next bound the number of actual subgroups of each type containing the fixed element.

Lemma 1. *The odd-dimensional overgroup and the plus decomposition overgroup are unique. In minus dimension there is at most one extension field normalizer for each prime r, and at most one unitary normalizer. In plus dimension there are at most two unitary normalizers.*

Proof. The distinct primary spaces of X determine the line or the two negative summands uniquely. This proves the first assertion.

Suppose X lies in an extension field normalizer with scalar field $K = \mathbb{F}_{q^r}$. Its image in the field automorphism quotient has order dividing r, so X^r is K-linear. Its order is at least $(q^m + 1)/(2r)$. Since $r \leq m/2$ and $m \geq 4$,

$$\frac{q^m + 1}{2r} \geq \frac{q^m + 1}{m} > q^{m/3} + 1. \tag{4}$$

For completeness, factor $q^m + 1 = (q^{m/3} + 1)(q^{2m/3} - q^{m/3} + 1)$. The second factor increases with q. For $q = 2, m \geq 4$, put $a = 2^{m/3} > 5/2$; its value $a^2 - a + 1$ exceeds four at $m = 4$, and its derivative in m is $(\log 2)(2a^2 - a)/3 > 1$ throughout the range. Thus the second factor exceeds m, proving (4).

The acquired degree lemma for divisors of $q^m + 1$ now makes X^r irreducible. Thus $\mathbb{F}_q[X^r] = E$, a field of order q^{2m}, and $K \subseteq C_{\mathrm{End}_{\mathbb{F}_q}(V)}(X^r) = E$. There is only one subfield of that order. Moreover its transferred quadratic form is unique. Indeed, if $Q(v) = \mathrm{Tr}_{K/\mathbb{F}_q}(Q_K(v))$, then the values $Q(av)$ for all $a \in K$ determine $Q_K(v)$ by trace nondegeneracy; in characteristic two the squaring map on K is bijective, and in odd characteristic the squares span K additively. Consequently the full normalizer of this field and form is unique. Taking its intersection with the fixed ambient group cannot create a second maximal subgroup.

For a minus unitary normalizer, X^2 centralizes its quadratic scalar field. Its order is at least $(q^m + 1)/4$ in odd characteristic and the full order $q^m + 1$ in even characteristic. The same acquired degree lemma, and $(q^m + 1)/4 > q^{m/3} + 1$ for odd $q \geq 3, m \geq 3$, identify the scalar field uniquely inside E. The hermitian form is recovered from the ground-field polar form by trace pairing, so its normalizer is unique too.

For a plus unitary normalizer, X^2 is irreducible on the large summand, by that argument with $m - 1$. Its quadratic scalar field is therefore the unique quadratic subfield of the large endomorphism field. In particular X, which is a multiplier there, centralizes this field. The field automorphism induced by an element of the global unitary normalizer is the same on every summand; it must therefore be trivial everywhere. On the two-space X itself, rather than X^2, determines the quadratic scalar field, since h is primitive of order $q + 1 > 2$. This covers $q = 3$ too. There are two possible relative identifications of the two quadratic fields, modulo a simultaneous field automorphism. For each the polar trace pairing determines the hermitian form. Thus at most two normalizers occur. These are bounds on actual field structures and forms, not just their conjugacy classes. □

Lemma 2. *In all three families, $|N_G(\langle x \rangle)| \leq 4m|T|$.*

Proof. Conjugating x to a generator of its cyclic group conjugates X to a scalar multiple of a coprime power of X. It therefore preserves the algebra generated by X, and its primary spaces. On an irreducible $2k$-space a normalizer of the field algebra has at most $2k$ possible Frobenius actions, each with fibre its multiplier group. There is one such block in the first two families, and in plus type there are at most $2(m - 1)$ choices on the large block and two on the small block. The conformal condition and division by common scalars give the fibre T of (3). These bounds are at most $4m|T|$. Only elements of G are counted, so triality automorphisms do not enlarge this normalizer. □

Lemma 3. *The actual overgroups in Lemma 1 have total ν bounded by the applicable terms in the following list:*

$$\begin{array}{ll} \textit{odd-dimensional negative overgroup} & 6q^{m^2}, \\ \textit{plus decomposition} & 12(q+1)q^{(m-1)^2}, \\ \textit{all unitary normalizers} & 8(q+1)q^{(m^2+m-2)/2}, \\ \textit{all minus extension field normalizers} & 8mq^{\lfloor m^2/2 \rfloor}. \end{array} \tag{5}$$

Proof. The acquired conformal bounds give $\nu(\mathrm{PCO}_d(q)) \le 3q^{\lfloor d^2/4 \rfloor}$ for odd q, and the corresponding bound with constant four for even q. For even q the latter follows from the retained automorphism involution estimate [2, Proposition 1.3], including the separate small derived-group identifications. In dimension two the negative conformal group is dihedral, with $\nu = q+3$ for odd q and $\nu = q+2$ for even q; both are at most $2(q+1)$.

Restriction to the negative hyperplane in odd total dimension has projective kernel of order at most two, giving $6q^{m^2}$ by (1). The decomposition subgroup maps into the product of the conformal quotients on its summands; its kernel has order at most two for odd q, and one for even q. This gives respectively $12(q+1)q^{(m-1)^2}$ and the smaller $8(q+1)q^{(m-1)^2}$.

Each unitary normalizer maps to $\mathrm{Aut}(\mathrm{PSU}_m(q))$ with scalar kernel of size at most $q+1$. Its simple derived group is in the valid range $m \ge 4$, including the usual small isomorphisms. The retained automorphism estimate, with the identity added, is at most $4q^{(m^2+m-2)/2}$. Lemma 1 allows at most two normalizers, giving the third term.

An extension field orthogonal normalizer acts on the corresponding derived projective group over q^r, with kernel of order at most two. The same estimate is at most $4q^{m^2/r}$; in dimension four minus type use $P\Omega_4^-(q^r) \cong \mathrm{PSL}_2(q^{2r})$. The field automorphisms are already included. There are at most m allowable primes r, and $r \ge 2$, giving the last bound. Faithfulness modulo the indicated scalar kernels is the standard natural-action interface of these classical groups; it is retained explicitly, rather than proved by a bound on their orders. □

3 Natural reversing automorphisms

The torus proof excludes nontrivial ordinary field involutions inverting x. Outside plus dimension eight the remaining diagram automorphisms are represented on the natural module. Quotients of two such inverters centralize x, so there are at most $|T|$ of them, whether that coset is inner-diagonal or the other natural orthogonal coset. The acquired odd-characteristic fixed-group estimates are

$$\nu(C_G(\alpha)) \le \begin{cases} 18q^{m(m+1)/2}, & \dim V = 2m+1, \\ 24q^{2\lfloor m^2/4 \rfloor}, & \dim V = 2m. \end{cases} \tag{6}$$

They include conformal lifts of order four and nonsquare multipliers.

For even q in dimension $2m$, the inverter normalizes to an isometry of rank m with nonalternating induced form. The accepted exact centralizer sequence has kernel $q^{m(m-1)/2}$ and quotient $\mathrm{Sp}_{m-1}(q)$ for odd m, and kernel $q^{(m^2+m-2)/2}$ and quotient $\mathrm{Sp}_{m-2}(q)$ for even m. Thus

$$\nu(C_G(\alpha)) \le 4q^{e_m}, \qquad e_m = \begin{cases} (3m^2-2m-1)/4, & m \text{ odd}, \\ (3m^2-4)/4, & m \text{ even}. \end{cases} \tag{7}$$

The small quotient $\mathrm{Sp}_4(2)$ has $\nu = 76$, below the bound used; it is not treated as simple. For $m = 4$, the exact quotient $\mathrm{Sp}_2(q)$ has $\nu = q^2$, so the sharper bound is q^{11}.

4 The uniform inequalities

The acquired involution lower bounds are $q^{m^2+m}/4$ in odd dimension and $q^{m^2-1}/4$ in even dimension. All are strict.

Proposition 4. *Criterion* (2) *holds in odd dimension* $2m+1$ *outside* $(m,q)=(3,3)$*, and in both even-dimensional families for* $m \geq 5$*, outside* $(m,q)=(5,2),(6,2)$*.*

Proof. In odd dimension, division by q^{m^2+m} bounds the right side by

$$18(1+q^{-m})q^{-m(m-1)/2} + 6q^{-m} + 4m(1+q^{-m})q^{-m^2}. \tag{8}$$

For $m=3, q \geq 5$ its maximum is at $q=5$ and is less than $194/1000$. For $m \geq 4, q \geq 3$ the maximum is at $(4,3)$ and is less than $1/10$. Each term decreases with q and with m in these ranges: for the last term the coefficient ratio $(m+1)/m$ is smaller than the lost factor q^{2m+1}. Both maxima are below $1/4$.

For odd q and $m \geq 5$, put $D = m^2-1$. We have $|T| \leq 3q^m/2$. Using Lemma 3 as an overcount for either sign, the normalized right side is at most

$$36q^{m+2\lfloor m^2/4\rfloor - D} + 16q^{m^2-2m+2-D} + 11q^{(m^2+m)/2-D} + 8mq^{\lfloor m^2/2\rfloor - D} + 6mq^{m-D}. \tag{9}$$

At $(m,q)=(5,3)$ this is less than $25/1000 < 1/4$. The successive gaps in the first exponent are $7, 11, 17, 23, \ldots$; the remaining gaps are $2m-3$, $(m^2-m-2)/2$, $\lceil m^2/2\rceil - 1$, and m^2-m-1. All increase, and the gap increases in the terms with coefficient m more than pay the factor $(m+1)/m$. Thus every term decreases.

For even q, use $|T| \leq 2q^m$. The corresponding upper bound is

$$8q^{e_m+m-D} + 18q^{m^2-2m+2-D} + 12q^{(m^2+m)/2-D} + 8mq^{\lfloor m^2/2\rfloor - D} + 8mq^{m-D}. \tag{10}$$

The first exponent gap $D-m-e_m$ equals k^2-1 at $m=2k+1$ and $k(k-2)$ at $m=2k$. It is three for $m=5,6$, eight for $m=7,8$, and increases in consecutive odd-even pairs thereafter. At $(5,4)$ and $(6,4)$ the total in (10) is less than $127/1000$. For $m \geq 7, q \geq 2$ it is maximized at $(7,2)$, where it is less than $41/1000$. The other exponent gaps increase as in (9), again paying the coefficients m. All these bounds are below $1/4$. The omitted even-field parameters are in the acquired finite list. □

5 Dimension eight

The acquired odd-characteristic subspace counts give $i_2(P\Omega_8^+(q)) > q^{16}/4$ and $i_2(P\Omega_8^-(q)) > q^{16}/2$. Here is the even-characteristic substitute.

Lemma 5. *For even* q*, both* $\Omega_8^+(q)$ *and* $\Omega_8^-(q)$ *have more than* $q^{16}/2$ *involutions.*

Proof. On $\mathbb{F}_q^4 \oplus \mathbb{F}_q^4$ use

$$Q_b(x,y) = \Big(\sum_i x_i\Big)^2 + x^T y + \sum_i b_i y_i^2, \qquad t(x,y) = (x+y, y).$$

The polar form is nondegenerate. Its Arf class is that of $\sum b_i$; choosing its absolute trace to be zero or one gives the two types. The map t preserves Q_b, has rank four, and has Dickson invariant zero, so belongs to Ω. The already proved centralizer formula gives $|C_\Omega(t)| = q^9 |\operatorname{Sp}_2(q)|$. One actual Ω conjugacy class therefore has size

$$q^2(q^4-1)^2(q^6-1) \quad \text{or} \quad q^2(q^8-1)(q^6-1),$$

respectively. The first ratio to q^{16} is at least $(15/16)^2(63/64) > 1/2$; the second is larger. This counts one actual class and needs no assertion about fusion of orthogonal classes. □

In minus dimension eight the only extension prime in the cover is $r = 2$, and there is no unitary type. For odd $q \geq 3$ the bound

$$24(q^4+1)q^8 + 8q^8 + 16(q^4+1) < q^{16}/2 \tag{11}$$

follows by dividing by q^{16}: the resulting decreasing function is less than $302/1000$ at $q = 3$.

For even $q \geq 4$, write $T_+ = (q^3+1)(q+1)$. The bound

$$q^{11}T_+ + 20(q+1)q^9 + 32q^8 + 16T_+ < q^{16}/2 \tag{12}$$

holds since the ratio is decreasing and is less than $324/1000$ at $q = 4$. It pays minus type already; in plus type it pays the natural decomposition and two unitary normalizers, if they form the complete cover after excluding the line stabilizers by the triality argument.

More precisely, the following last two bounds are conditional on that complete plus-dimensional cover. The separately reviewed triality argument chooses the primitive components so that for odd $q \geq 7$ there are no line overgroups in any of the three eight-dimensional representations, and at most T_+ inverters. Then

$$24T_+q^8 + 20(q+1)q^9 + 16T_+ < q^{16}/4, \tag{13}$$

as the decreasing ratio is less than $12/1000$ at $q = 7$. For $q = 5$ there are at most two inverter cosets and six actual $\Omega_7(q)$ line stabilizers, each with $\nu \leq 3q^{12}$. Consequently

$$48T_+q^8 + 18q^{12} + 20(q+1)q^9 + 16T_+ < q^{16}/4, \tag{14}$$

whose ratio at five is less than $124/1000$. The cases $q = 2, 3$ in plus type and $q = 2$ in minus type have acquired finite witnesses. No multiplicity-two assumption for the $q = 5$ line overgroups is used. The missing complete-cover classification must still be supplied before (12)–(14) prove the plus dimension-eight generation theorem.

Attribution. The finite classical group orders, trace and form theory, natural automorphism actions, and the quoted maximal subgroup classifications are borrowed. The actual-field multiplicity arguments, the consistent projective centralizers, the new fixed-group counts and the uniform inequalities are the contributions used here. Full generation outside the explicitly conditional case is relative to those named inputs; no newly solved conjecture, additional original, Lean result or numerical compression factor is claimed by this working note.

References

[1] D. Bubboloni, P. Spiga and Th. Weigel, *Normal 2-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Lemmas 6.1,7.3,8.1.

[2] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), 393–464, Proposition 1.3.

Choosing an orthogonal torus element to separate the triality modules

Human Compression project

This draft supplies the triality interface in plus dimension eight. It awaits distinct review and does not classify all maximal overgroups.

Let $G = P\Omega_8^+(q)$, with inner-diagonal automorphism group J. We retain the standard automorphism structure: modulo J, the diagram group S_3 commutes with the cyclic field group. The three diagram points are the natural and two half-spin eight-dimensional projective representations. The stabilizer of one point acts by natural projective conformal maps in that representation. The Clifford and triality descriptions work in characteristic two as well; see [1, Sections 3.1 and 3.4], in particular Theorem 3.12 and the isomorphisms (3.20)–(3.21). These are borrowed structural statements.

Put $A = q^3 + 1$, $B = q + 1$, $b = q^2 - q + 1$, so $A = Bb$. Choose g of order A in the norm-one torus of $\mathbb{F}_{q^6}/\mathbb{F}_{q^3}$. On negative spaces of dimensions six and two use primitive multipliers g and $h = g^{ub}$, with u a unit modulo B. Let X be their direct sum and x its image in G. The accepted torus proof gives $X \in \Omega_8^+(q)$, $|x| = A/\gcd(2, q-1)$, and the connected adjoint centralizer

$$C_J(x) = T, \qquad |T| = AB. \tag{1}$$

In particular its order is not replaced by $\gcd(2, q-1)|x|$.

Lemma 1. *Modulo scalar signs, the natural eigenvalue exponents relative to g are*

$$\pm 1, \quad \pm q, \quad \pm q^2, \quad \pm ub.$$

The two half-spin lists are

$$c_\epsilon q^j \ (0 \le j < 6), \quad d_\epsilon, -d_\epsilon, \qquad c_\epsilon = \frac{1 + q + q^2 + \epsilon ub}{2}, \quad d_\epsilon = \frac{(1 - \epsilon u)b}{2}, \quad \epsilon = \pm 1. \tag{2}$$

For even q, division by two is modulo A. For odd q the numerators are even integers, and comparison modulo scalar signs is modulo $A/2$. Changing a Spin lift changes only a scalar sign.

Proof. The Clifford construction on the exterior algebra of an isotropic four-space gives the weights $\prod z_i^{s_i/2}$, $s_i = \pm 1$, when the natural eigenvalues are z_i, z_i^{-1}. The two parities of the number of minus signs give the two half-spin modules. Indeed the ith torus factor acts with weight $z_i^{1/2}$ on exterior monomials containing the ith basis vector and with weight $z_i^{-1/2}$ on the others. Conjugation on the creation and contraction operators then has the required weights z_i, z_i^{-1}. Thus here the weights are the half-sums of $1, q, q^2, ub$ in the two parities.

Use $q^3 = -1$ and $qb = -b$ modulo A. Successively multiplying the all-positive half-sum by q gives six even-parity half-sums; the other two are $\pm(1 - u)b/2$. Starting with $1 + q + q^2 - ub$ gives the odd-parity list. For odd q the doubled identities modulo A prove the desired identities modulo $A/2$. For even q, two is invertible modulo A. □

For $q \ge 3$, the order of q modulo b is six. It divides six because $q^3 = -1$; $b > q + 1 > 2$ excludes orders one and three, and $q^2 = 1$ would force $q = 2 \pmod b$, excluding order two. Moreover

$$c_\epsilon \equiv q \pmod b. \tag{3}$$

Here $1 + q + q^2 \equiv 2q \pmod b$ and b is odd.

Proposition 2. *Suppose q is even and $q \geq 4$, or q is odd and $q \geq 7$. Choose a unit $u \pmod B$ other than 1 and -1. Every involutory automorphism of G inverting x belongs to J. There are at most AB such automorphisms. Furthermore, x belongs to no triality image of a stabilizer of a nonsingular line in a natural eight-dimensional module.*

Proof. For even q take $u = 2$. For odd $q \geq 7$, B is even and at least eight, and $\varphi(B) > 2$. The prime-power formula for Euler's function shows that $\varphi(n) = 2$ only for $n = 3, 4, 6$.

Put $M = A$ for even q and $M = A/2$ for odd q. By (3), the six exponents $c_\epsilon q^j$ have period six already modulo b; the two exponents $\pm d_\epsilon$ are divisible by b and cannot join that orbit. A similarity up to scalar signs between a half-spin and the natural spectrum must therefore match their six-element orbits. It implies $c_\epsilon = q^j \pmod M$ for some j. Reduction modulo b forces $j = 1 \pmod 6$. Since

$$2(c_\epsilon - q) = (1 + \epsilon u)b,$$

the congruence implies $u = -\epsilon \pmod B$, a contradiction. Taking inverses does not change either spectrum, so this also excludes a diagram transposition moving the natural module for an inverter.

A nontrivial involutory field projection has $q = p^f$ with even f and exponent $f/2$. If its diagram part moves the natural module, the same comparison, reduced modulo b, gives $p^{f/2}q = q^j$ after absorbing inversion into the six-element orbit. Squaring gives $q^3 = q^{2j}$, impossible for an element of order six. Ordinary natural field inverters were excluded in the acquired torus proof. An involution cannot project to a three-cycle in S_3.

The remaining diagram transposition fixes the natural module. Apply q^3-Frobenius on the six-space and q-Frobenius on the two-space. This is an involutory isometry inverting X. It is proper: its determinant in odd characteristic is $(-1)^3(-1) = 1$, and its Dickson invariant in characteristic two is $3 + 1 = 0 \pmod 2$. It therefore represents an element of J. Any other natural inverter differs from it by an element of the natural projective centralizer, the connected torus T. Thus no improper natural inverter exists. All inverters lie in J, where their pairwise quotients centralize x, proving the bound $|T|$.

Finally the natural module has no scalar-sign eigenvalue. In either half-spin module a large-orbit exponent is nonzero modulo b. A small exponent is zero modulo M only if $u = \epsilon \pmod B$, also excluded. There is no scalar-sign eigenvalue in any module. A nonsingular line fixed by a projective isometry has eigenvalue 1 or -1 on an isometric lift, since its nonzero quadratic value is preserved. Such a line therefore does not exist. □

Proposition 3. *For $q = 5$ take $u = 1$. At most $2AB$ involutory automorphisms invert x. Their fixed subgroups in G each have at most $24q^8$ elements of square one, including the identity. There are at most six triality images of nonsingular-line stabilizers containing x.*

Proof. Here $A = 126$, $B = 6$, $b = 21$, and $M = 63$. Formula (2) gives one half-spin spectrum identical modulo signs to the natural one. The other has large-orbit representative 26, of order 63, and a scalar two-space. The natural and first half-spin modules have no scalar-sign eigenvalue, so the remaining module is intrinsically distinguished. An inverter must fix its diagram point. There are at most two possible diagram projections and no field automorphism. Each such coset of J has at most $|T|$ inverters.

Use the distinguished module to represent an inverter by a conformal map a. Its six-space has irreducible eigenvalue order 63. Projective inversion has scalar 1: the degree argument forces the field exponent to be three, already giving exact inversion. On that block, the square and conformal multiplier of a are the same scalar c, as in the natural torus proof. Since its projective action is involutory, $a^2 = cI$ on the entire module.

If c is a square, the six-space contributes eigenspace dimensions three and three. The two-space adds $2, 0$, $1, 1$, or $0, 2$. Thus the full pair is $(5, 3)$, $(4, 4)$, or $(3, 5)$. The written odd-characteristic projective fixed-group bounds give $18q^8$ in the unequal cases and $24q^8$ in the

balanced case. If c is nonsquare, the two eigenspaces over the algebraic closure each have dimension four. The same bound applies, including the quadratic-field form and projective kernel.

Only the distinguished module can have a fixed nonsingular line. Its scalar two-space has minus type: the irreducible six-space is negative and the eight-space is positive. Its $q + 1 = 6$ lines are all nonsingular, and they are the only possible lines. There is at most one stabilizer for each line. This counts actual stabilizers, without replacing their multiplicity by the number of conjugacy types. □

Attribution and limits. The three representations and triality action, finite automorphism structure and Clifford construction are prior theory. The exponent calculation, primitive choice separating spectra, and resulting coset and actual-line bounds are the arguments here. The cases $q = 2, 3$ have acquired finite witnesses. The separate maximal-overgroup cover and final generation inequality remain necessary; no whole-original or new-conjecture claim is made.

References

[1] D. Barry and J.-P. Tignol, *Trialitarian triples*, arXiv:2301.10130v1, Sections 3.1–3.4. This auxiliary source does not enlarge the fixed original catalogue.

[2] D. Bubboloni, P. Spiga and Th. Weigel, *Normal* 2*-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Section 8, physical pages 81–82 for the triality-related orthogonal line stabilizers.